

MEMS Resonators for Frequency Reference and Timing Applications

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Abstract—An overview of microelectromechanical systems (MEMS) resonators for frequency reference and timing applications is presented. The progress made in the past few decades in design, modeling, fabrication and packaging of MEMS resonators is summarized. In particular, the state-of-the-art technologies for improving the overall performance of MEMS resonators, such as quality factor (Q), motional impedance, temperature sensitivity, and initial frequency uniformity, are reviewed in detail. The challenges and opportunities during the commercialization of MEMS resonators are also stated, and future development trends driven either by technology or market are outlined. This paper intends to provide an outlook for possible research directions of MEMS resonators in frequency reference and timing applications. With outstanding reliability, unique multi-frequency functionality on a single chip, and high accuracy, MEMS resonators show great potential for replacing the quartz crystal resonators which have been dominating the timing market since 1920s. [2020-0106]

Index Terms—Resonators, oscillators, frequency reference, timing, microelectromechanical systems.

I. INTRODUCTION

RESONATORS are the most essential components for building reference oscillators, which are the ubiquitous elements used in the electronics industry and constitute a multi-billion dollar market [1]–[3]. Mechanical and electrical oscillators are two separate technology families for oscillators. In electrical oscillators, the frequency-selective element comprises a resistor-capacitor (RC) or inductor-capacitor (LC) filter and is integrated on an electronic circuit to produce a periodic, oscillating electronic signal [4]–[6], often a sine wave or a square wave. Mechanical oscillators are typically based on mechanical resonators as the frequency-selective element, which demonstrate a superior frequency stability and noise performance compared to electrical oscillators, since

the mechanical resonance exhibits a much higher quality factor (Q) than the electrical resonance.

Mechanical oscillators involve moving mechanical resonant elements, actuated or sensed with electrical signals. Quartz crystal oscillators and microelectromechanical system (MEMS) oscillators are two main categories for mechanical oscillators. Quartz crystal oscillators were developed as early as in 1920s and have since dominated the overall timing market. MEMS oscillators have experienced rapid growth, and have demonstrated rivaling electrical performance with the well-established quartz crystal oscillators recently. According to Yole Développement, the MEMS based oscillator market is valued at USD 99 million in 2018 and is expected to reach USD 604 million by 2024, with a compound annual growth rate (CAGR) of 38% [7].

As the frequency-selective element, the mechanical resonators are the most important component in mechanical oscillators. In recent years, MEMS resonators have shown great potential to replace quartz crystal resonators due to their compact size, best reliability, high temperature stability, and good compatibility with the complementary metal-oxide-semiconductor (CMOS) standard fabrication process [8]–[14]. Although the quartz crystal resonators have served the electronics industry well for almost 100 years and are still dominating the whole timing market, they suffer from several disadvantages compared with the MEMS resonators [15], [16].

One of the major advantages of MEMS resonators over quartz crystal resonators is the fabrication process. Quartz crystal resonators are typically individually cut using traditional precision manufacturing technologies, while MEMS resonators are batch fabricated on wafer substrates using semiconductor processes [1], [3]. Leveraging semiconductor processes, MEMS resonators show higher reliability than quartz crystal resonators [17]–[19]. Failure modes of quartz crystal resonators include poor electrical connection, oscillation die damage, quartz contamination, defects in quartz welding process, and corrosion [20]. Secondly, MEMS resonators are more immune to external environmental noises, such as vibrations, shocks, and electromagnetic interferences (EMI), which show better reliability than quartz crystal resonators [17]–[19]. Thirdly, multi-frequency resonators can be integrated on a single chip for MEMS resonator, thanks to the flexibility in resonator designs using MEMS technologies. While quartz crystal resonators have strict design rules, especially for classical thickness shear mode quartz crystal resonators, the resonant frequency is determined by the

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TABLE I
COMPARISON OF TYPICAL SPECIFICATION OF MHz QUARTZ
CRYSTAL OSCILLATORS AND MEMS OSCILLATORS

Specification	Quartz crystal oscillator (TCXO)	MEMS oscillator (TCXO)
Resonator technology	Quartz crystal	MEMS
Manufacturing technology	Traditional precision manufacturing	Micromachining
Multi-frequency functionality on a single chip	No	Yes
Packaging form	Die-level	Wafer-level
Accuracy (ppm)	5	10
Temperature stability (ppm)	2	5
Reliability (mean time between failures) (hr)	1×10^7	5×10^8
First-year aging (ppm)	± 5	± 2
Vibration sensitivity (ppb/g)	3	0.1
Size (mm ³)	50	5
Unit price, typical, (USD)	~ 1	< 1

thickness of the piezoelectric layer, which hinders achieving multi-frequency functionality on a single chip. Moreover, MEMS resonators have smaller size and can easily achieve higher natural resonant frequencies without using higher-order vibrating modes [3]. In summary, the pros and cons of present day quartz crystal oscillators and MEMS oscillators can be schematically depicted in Table I. Part of specifications of Table I are adapted from [3] and updated based on [17]–[19], [21], [22]. It is seen that MEMS oscillator is currently a viable replacement for quartz TCXO (temperature-compensated crystal oscillator), which is similar in terms of performance, but with a much smaller size, lower long-term aging and much higher reliability.

MEMS resonators are based on mechanically resonating micro-structures, which were introduced as a resonant gate transistor in their early form in 1967 [23]. They have gained significant research and commercial interest due to their wide and high impact applications, such as sensing, filtering, frequency reference and timing applications. In sensing applications, changes in the resonant frequency of MEMS resonator are used to monitor a given quantity (temperature [24], [25], charge [26], [27], pressure [28], [29], acceleration [30], flow [31] and gas concentrations [32], [33], etc.). In filtering applications, resonant structures are employed to implement filters that can be used in radio frequency wireless transceivers [34]. In timing applications, MEMS resonators vibrating at resonance are used as the frequency-selective element within an electronic system to generate a high quality frequency or clock signal. This paper offers a review of MEMS resonators for frequency reference and timing applications.

The major performance metrics of MEMS resonators include resonant frequency (f_0), Q , motional resistance (R_m), temperature stability, initial frequency uniformity, electromechanical coupling coefficient (k_t^2), frequency- Q product ($f \times Q$), and aging. Additional aspects such as power handling capability, and vibration/shock sensitivity, may also be critical depending on application.

Resonant Frequency: f_0 is the natural vibration frequency of the MEMS resonators, which is determined by the physical

parameters and operating conditions of the MEMS resonators. Due to the miniature size, the f_0 of MEMS resonators are typically in kHz to MHz range, but can be extended to the GHz range with appropriate design.

Quality Factor: Q is perhaps one of the most important figures of merit for a resonant system. In physics, Q is defined as the ratio of the total stored energy to the average energy loss per cycle. The commonly encountered energy loss mechanisms for MEMS resonators include anchor loss (Q_{anc}), thermoelastic dissipation (Q_{TED}), air damping (Q_{air}), and intrinsic material loss (Q_{mat}). The overall Q is determined by the sum of all energy losses from each energy dissipation mechanism. For MEMS resonators, a higher Q is generally desired for most applications. Higher Q corresponds to realizing a low phase noise oscillator [34], as well as achieving high-resolution resonant sensors.

Motional Impedance: R_m captures the net equivalent electrical energy loss for a MEMS resonator when it is vibrating at the resonant frequency. For MEMS resonators, R_m is inversely proportional to the mechanical Q and the effective electromechanical coupling coefficient of the MEMS resonator. A low R_m is desired for a MEMS resonator in order to reduce the power consumption and improve the signal-to-noise ratio as the resonator is vibrating in a sustained oscillation loop.

Temperature Stability: For clock signals in timing applications, time or reference frequency is desired to be constant. However, in reality, MEMS resonators show frequency variations due to changes in temperature. Changes in temperature result in corresponding changes in the materials properties such as the elastic moduli, and the geometrical thermal expansion, thereby changing the effective stiffness and geometry of the resonator and thus also the resonant frequency. The sensitivity of a resonator's frequency to temperature changes is defined by the temperature coefficient of frequency (TCF), which for silicon is typically in the range of -20 to -35 ppm/ $^{\circ}\text{C}$. A low TCF is required to achieve a high stable timing reference.

Initial Frequency Uniformity: In timing applications, the frequency output of MEMS oscillators should not only be accurate, but also be precise. Accuracy describes how close the frequency is to the desired value. While precision or uniformity represents how well all the resonators on a wafer, or across wafers, match each other in terms of the output frequency. Thus, as the frequency-selective element, MEMS resonators are required to have accurate and precise control of the predefined initial resonant frequencies. The initial frequency uniformity is one of the most important parameters for MEMS resonators, and determines the yield and manufacturability of these devices. Achieving part-per-million (ppm)-level frequency accuracy and uniformity in batches of MEMS resonators is a daunting task necessitating post-fabrication trimming.

Electromechanical Coupling Coefficient: k_t^2 is related to the conversion rate between mechanical energy and electrical energy, which is defined as the ratio of the output mechanical energy and the input electrical energy in MEMS resonators. This is different from the turn ratio or electromechanical

coupling factor, η , defined in Eq. (3), which relates to the conversion from voltage to force or from charge to displacement. A MEMS piezoelectric resonator usually demonstrates much higher k_t^2 than a MEMS capacitive resonator.

$f \times Q$: For a given material, $f \times Q$ often presents a fundamental upper limit due to the effect of material-limited intrinsic dissipation mechanisms. In silicon MEMS resonators, the dissipation due to phonon-phonon interactions, in particular Akhieser dissipation, is the dominant intrinsic loss mechanism.

Aging: Resonator's frequency tends to drift over time, due to changes of material or stress in the resonator or the packaging over time. Long-term aging represents the long-term frequency stability of resonator, which is particularly important when utilized in frequency reference and timing applications. Long-term stability is usually measured in months to years. First-year aging and 10-year aging are the most commonly used parameters.

Following this section, the operation principle and designs of MEMS resonators are reviewed in section II. The details of the fabrication and packaging technologies of MEMS resonators are described in section III. Then, the key technologies for developing high performance MEMS resonators are presented in section IV. Section V illustrates the business activities of the major players in the MEMS resonator market. The future trends and opportunities are provided in section VI. Section VII offers the outlook for the developments of MEMS resonators and finally section VIII concludes the paper.

II. OPERATION PRINCIPLE AND DESIGNS

MEMS resonators are electrically excited into resonance, with electro-mechanical transduction mechanisms varying from capacitive, piezoelectric, thermal and others. The most commonly used transduction mechanisms are piezoelectric or capacitive.

A. Mechanical and Electrical Models

In the mechanical domain, the dynamic mechanical behavior of a MEMS resonator can be represented by a mass-damper-spring system, as shown in Fig. 1. The kinetic and potential energies of the MEMS resonators are continuously converted to each other. The relationship between the displacement of the mass, x , and external excitation force, $f_{ex}(t)$, can be expressed as a second order vibration system:

$$m_{eq} \frac{\partial^2 x}{\partial t^2} + c_{eq} \frac{\partial x}{\partial t} + k_{eq} x = f_{ex}(t) \quad (1)$$

where m_{eq} , c_{eq} , and k_{eq} are the effective mass, effective damping coefficient, and effective stiffness in the system, respectively.

The system transfer function, $H(\omega)$, of the mass-damper-spring system can be obtained by taking the Fourier transform of Eq.(1) and is given by:

$$H(\omega) = \frac{X(\omega)}{F_{ex}(\omega)} = \frac{1}{k_{eq}} \frac{\omega_0^2}{\omega_0^2 - \omega^2 + j \frac{\omega \omega_0}{Q}} \quad (2)$$

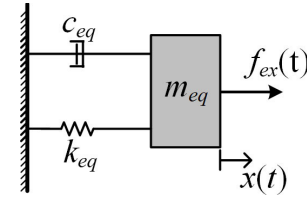


Fig. 1. In the mechanical domain, a MEMS resonator can be modeled by a mass-damper-spring system.

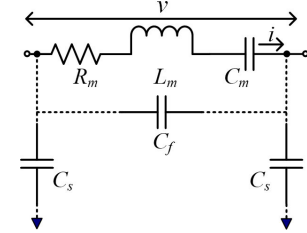


Fig. 2. In the electrical domain, a MEMS resonator can be modeled as an equivalent resistance R_m , inductance L_m , and capacitance C_m connected in series.

where ω is the angular frequency, $\omega_0 = \sqrt{k_{eq}/m_{eq}}$ denotes the natural resonant frequency, and $Q \approx k_{eq}/(c_{eq}\omega_0)$ presents the mechanical quality factor of the system. The uppercase letters in Eq.(2) represent Fourier transformed variables.

In the electrical domain, the MEMS resonator is modeled as an equivalent resistance R_m , inductance L_m , and capacitance C_m connected in series, as shown in Fig. 2. We assume that i is the ac current going through the resonator and v is the ac voltage drop across the two terminals of the resonator. The transduction from the electrical to mechanical domains and vice versa are modeled with transformers with proper winding ratios of controlled voltage or current sources. The turn ratio, or electromechanical coupling factor, $\eta = i/\frac{\partial x}{\partial t}$ is defined as the ratio of the electrical current in the electrical domain and the mechanical velocity in the mechanical domain.

Considering that the power dissipated in the resonator can be written as the product $v \cdot i$ in the electrical domain and as $f_{ex} \cdot \frac{\partial x}{\partial t}$ in the mechanical domain results in:

$$f_{ex} = \eta v \quad (3)$$

Combining Eq.(1)-(3), the equation of motion of the resonator can be expressed in terms of alternating current and voltage:

$$L_m \frac{\partial i}{\partial t} + R_m i + \frac{1}{C_m} \int i \partial t = v \quad (4)$$

in which, L_m , R_m , and C_m are expressed as:

$$L_m = \frac{m_{eq}}{\eta^2}, \quad R_m = \frac{c_{eq}}{\eta^2}, \quad C_m = \frac{\eta^2}{k_{eq}} \quad (5)$$

Obviously, Eq.(1) describes the mechanical motion of the mass-damper-spring system in the mechanical domain and Eq.(4) describes the electrical current motion of the R_m - L_m - C_m branch in the electrical domain. Both of these two equations are second-order systems, and have a similar form. The equivalent electrical circuit allows for co-simulation of

TABLE II
MAPPING OF THE QUANTITIES BETWEEN THE
ELECTRICAL AND MECHANICAL DOMAINS

Mechanical domain	Electrical domain
Force, f	Voltage, v
Velocity, $\frac{\partial x}{\partial t}$	Current, i
Displacement, x	Charge, q
Compliance, $1/k_{eq}$	Capacitance, C_m
Mass, m_{eq}	Inductance, L_m
Damping, c_{eq}	Resistance, R_m

the entire system within the electrical domain. The quantities in the mechanical domain have corresponding quantities in the electrical domain, in order to represent the mechanical device with electrical elements. Table II summarizes the common set of mapping rules [8], [35].

In most cases, a feedthrough capacitance C_f is also present between the two resonator terminals in parallel to the series R_m - L_m - C_m branch. The feedthrough capacitance is caused either by the dielectric permittivity of the resonator itself or by on-chip and off-chip parasitic capacitances, or both. The feedthrough capacitance C_f may be shunted to ground as shunt capacitance C_s instead where possible, to provide a steeper phase transition for better noise performance, at the expense of power consumption.

B. Vibration Modes

Mechanical resonators can be broadly classified into several general categories based on their mode of operation. The common operation modes include flexural, torsional, bulk, and surface modes [8], [36], [37].

The typical surface mode devices are surface acoustic wave (SAW) resonators, in which a SAW (most commonly Rayleigh wave) is generated using interdigital transducers (IDTs) on a piezoelectric substrate to excite harmonic vibrations that decay from the surface into the bulk [38]. Film-bulk acoustic resonator (FBAR) or solidly mounted resonator (SMR) is a typical bulk mode device, which consists of a piezoelectric material sandwiched between two electrodes and is acoustically isolated from the surrounding medium [39]–[41]. SAW resonators typically provide resonant frequencies ranging from several hundred MHz to few GHz and FBARs offer higher fundamental operation frequencies above few GHz. FBAR-based MEMS oscillator features high mechanical resonant frequency and high Q , resulting in excellent phase noise and jitter performance among MEMS oscillators [42]. Although, these two kinds of devices have been demonstrated in timing applications [42], [43], they are mostly used to build the filters in radio-frequency (RF) systems. Thus, FBAR and SAW resonators are not described much in this paper. Most of MEMS resonators used in frequency control and timing applications are operating in the flexural, torsional, and bulk modes, as shown in Table III.

The flexural mode resonators are characterized by the bending of long or thin geometries, such as a beam or a membrane, as shown in Table III. Vibrations in flexural mode are due to transverse standing waves. The principal stress is approximately aligned with the normal stress along the

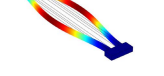
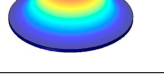
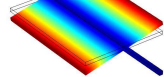
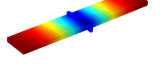
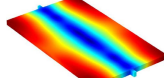
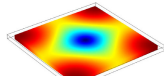
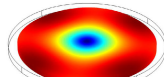
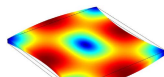
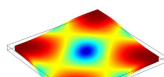
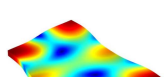
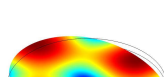
structure and is orthogonal to the linear displacement of the flexure. The vibration mode shape is determined by the resonator structure and boundary conditions applied. MEMS resonators with beam structures operating in flexural mode typically include cantilevers, clamped-clamped (C-C) or double clamped beams, free-free (F-F) beams, and double-ended tuning forks (DETF). For membrane structure, the edges of the membrane are fixed and the rest parts in the membrane move up and down with a maximum displacement in the center. The membrane structure could also be fixed at the center of the membrane and the maximum deflection occurs at the edges of the membrane [73].

In torsional mode, the resonators are most typically designed in the form of beams or paddles supported by a pair of tethers [54], [74], [75], as shown in Table III. The resonator oscillates by the twisting of beams or tethers with respect to the center line defined by the beams or the tethers. Compared to flexural mode resonators, the torsional mode resonators with similar dimension scales, have large effective stiffness, high resonant frequency, low anchor loss and negligible thermal elastic damping [37], [75]. Torsional mode resonators are particularly interesting for various sensing applications, such as electrometers [26], biosensors [76], angular rate sensors [77], and magnetometers [78].

The bulk mode resonators feature longitudinal standing waves or shear waves occurring within the volume. Compared to the bending of the structures in the flexural mode resonators, bulk mode resonators are characterized by deformation of the structure through planar expansions or contractions. Thus, the lateral or vertical physical dimensions, such as width, length, or thickness, determines the resonant frequency of the resonator. Bulk mode resonators have much higher effective stiffness than the flexural mode resonators with of similar dimension scales. Bulk mode resonators could be beams, rectangular plates, square plates, and circular disks, as shown in Table III. The beam resonators are usually excited to vibrate in the length-extensional (LE) mode. For rectangular plate resonators, the typically used longitudinal bulk mode includes LE or width-extensional (WE) modes. In case of square plate MEMS resonators, the commonly operated longitudinal bulk mode is square-extensional (SE) mode [79], [80]. The circular disk MEMS resonators can be driven into oscillation in the lateral bulk mode, which is referred as the radial breathing mode. In longitudinal bulk mode, the MEMS structure compresses and expands apart from the center, where minimum displacement occurs. The longitudinal bulk mode resonators are usually clamped at their quasi-nodal point in the center of structure, which minimizes the anchor losses and results in lower energy losses and higher Q , compared to other vibration mode resonators.

Shear bulk mode is similar to the longitudinal bulk mode, in which the acoustic wavelength is also determined only by the lateral features of the structure. Compared to longitudinal bulk modes, shear waves instead of longitudinal waves are propagating in the structures. Therefore, the effective stiffness is defined by the shear modulus of the structure rather than the Young's modulus. In shear bulk mode, the resonator structure shears with particle displacements parallel to the

TABLE III
COMMON MEMS RESONATOR STRUCTURES, OPERATING MODES, AND THEIR FREQUENCY EXPRESSIONS

Vibration modes	Resonator structures	Mode shapes	Resonant frequency	Reference	Parameters
Flexural mode	Cantilever		$f_0 = \frac{\alpha_n}{\sqrt{12}} \sqrt{\frac{E}{\rho}} \frac{h}{L^2}$	[44]–[46]	α_n : n^{th} mode coefficient, which is 3.516 and 22.034 for the first two modes E : Effective elasticity modulus ρ : Mass density h : Device thickness L : Device length β_n : Mode coefficient given by the n^{th} root of the equation $\cos(\beta_n) \cosh(\beta_n) = 1$, which is 4.73, 7.853 and 10.996 for the first three modes n : Mode number W : Device width R : Device radius σ : Poisson ratio G : Shear modulus w : Tether width c : Tether length K_l : Constant parameter, dependent on w/h . K_l equals 1/3 when $w/h > 20$ a : Paddle width b : Paddle length λ_n : n^{th} mode coefficient, which is 1.99, 5.17 and 8.42 for the first three modes C_0 : Constant frequency parameter κ_n : Frequency parameter dependent on the orientation of the silicon crystal. $\kappa_n=1.6002$ in $\langle 110 \rangle$ and $\kappa_n=1.4082$ in $\langle 100 \rangle$ orientation, respectively.
	Clamped-clamped beam		$f_0 = \frac{\beta_n^2}{2\pi\sqrt{12}} \sqrt{\frac{E}{\rho}} \frac{h}{L^2}$	[47]	
	Free-free beam		$f_0 = \frac{\beta_n^2}{2\pi\sqrt{12}} \sqrt{\frac{E}{\rho}} \frac{h}{L^2}$	[48]–[50]	
	Double-ended tuning fork		$f_0 = \frac{[(1+2n)\pi/2]^2}{2\pi\sqrt{12}} \sqrt{\frac{E}{\rho}} \frac{W}{L^2}$	[51], [52]	
	Circular membrane		$f_0 = \frac{10.22}{2\pi R^2 \sqrt{\rho h/D}}, D = \frac{Eh^3}{12(1-\sigma^2)}$	[53]	
Torsional mode	Paddle		$f_0 = \frac{1}{2\pi} \sqrt{\frac{24K_l G w h^2}{c(2cw^3 + ab^3)}}$	[54]–[56]	
Bulk mode	Rectangular plate (Length-extensional)		$f_0 = \frac{n}{2L} \sqrt{\frac{E}{\rho}}$	[11], [57], [58]	
	Rectangular plate (Width-extensional)		$f_0 = \frac{n}{2W} \sqrt{\frac{E}{\rho}}$		
	Square plate (Square-extensional)		$f_0 = \frac{1}{2L} \sqrt{\frac{E}{\rho(1-\sigma^2)}}$	[59], [60]	
	Circular plate (Radial breathing)		$f_0 = \frac{\lambda_n}{R} \sqrt{\frac{E}{\rho(1-\sigma^2)}}$	[61]–[63]	
	Square plate (Lamé)		$f_0 = \frac{1}{\sqrt{2}L} \sqrt{\frac{G}{\rho}}$	[14], [64]–[67]	
	Square plate (Face-shear)		$f_0 = \frac{C_0}{L} \sqrt{\frac{G}{\rho}}$	[68]	
	Rectangular plate (Width-shear)				
	Circular plate (Wine glass)		$f_0 = \frac{\kappa_n}{2\pi R} \sqrt{\frac{E}{\rho(1-\sigma^2)}}$	[69]–[72]	

Note: assuming the materials used are isotropic.

surface of the structure, in which the top surface and the bottom surface move in opposite directions without change of volume [81]. The classic AT-cut quartz resonators are commonly vibrating in the thickness shear (TS) mode [3]. For square plate MEMS resonators, the commonly used shear bulk modes include the Lamé and face shear (FS) modes [68]. In circular disk resonators, the shear bulk mode is commonly referred to as the wine glass mode [70]. The width shear (WS) mode has also been realized in piezoelectric actuated MEMS resonators [25]. Table III also shows the different shear bulk mode resonators. Compared to the longitudinal wave, the shear wave has some unique features, such as extremely low velocity in the viscous liquid. Therefore, shear bulk mode resonators are more suitable for sensing applications in liquid due to their ultra small acoustic impedance, which determines the large reflection coefficient at the solid-liquid interface.

C. Transduction Mechanisms

According to the transduction type, MEMS resonators are mainly classified into two categories: electrostatic and piezoelectric. Electrostatic MEMS resonators use an electrostatic field interacting with the movable resonant structure, while the piezoelectric MEMS resonators use piezoelectric actuation to convert mechanical motion and stress to an electrical signal, and vice versa. Besides electrostatic and piezoelectric, other transduction mechanisms, such as thermal-piezoresistive [82], magnetic [78], and optical [83], [84], can also be employed for inter-conversion of energy between mechanical and electrical domain in the MEMS resonators.

Electrostatic MEMS resonators are usually made using SCS or polysilicon, in which the resonant body only includes one or two materials. The simple structure makes them more robust and reliable due to their fewer failure sources. A distinguishing feature of electrostatic MEMS resonators is the high Q , normally one or two orders of magnitude higher than their piezoelectric counterpart. In addition, SCS-based MEMS resonators have low material aging, since SCS is almost a perfect single crystal material presenting a high potential resistance against aging. However, electrostatic MEMS resonators suffer from low electromechanical transduction efficiency and hence high motional impedance. To reduce the motional impedance, a large dc bias and very narrow capacitive transducer gaps as small as few hundred nanometers are required. A large dc voltage and narrow gaps may cause additional nonlinearity to the resonators and the narrow gaps require precise process control of the fabrication steps.

Piezoelectric MEMS resonators are commonly comprised of a thin film piezoelectric layer sandwiched between two metal electrodes [39], [40], which are sometimes stacked on top of a SCS device layer. The piezoelectric resonators have higher electromechanical coupling factor, which results in a lower motional impedance [85], [86]. No dc bias is necessary for piezoelectric resonators, which eliminates the frequency drifts caused by dc voltage variations. No narrow gaps are required, which makes the piezoelectric resonators easy to achieve batch manufacturing with good process uniformity, despite the fact that the piezoelectric resonators have a relatively lower Q

TABLE IV
COMPARISON BETWEEN ELECTROSTATIC AND
PIEZOELECTRIC MEMS RESONATORS

Technology	Electrostatic	Piezoelectric
Principle	Electrostatic actuation & capacitive sensing	Piezoelectric actuation & sensing
Transduction area	Small	Large
Bias voltage, V_{dc}	Large	N.A.
Q	Very high (hundred thousands)	High (tens of thousands)
Electromechanical coupling factor, η	$\nabla C \cdot V_{dc}$	$2d_{31}E_{eff}L^\ddagger$
Motional resistance, R_m	$\frac{\sqrt{k_{eq}m_{eq}}}{Q} \frac{g^4}{V_{dc}^2 \epsilon^2 A^2}^\dagger$	$\frac{\pi}{8} \frac{T}{L} \frac{\sqrt{\rho_{eff}}}{QE_{eff}^2 d_{31}^2}^\ddagger$

where ∇C is the gradient of capacitance between two electrodes. d_{31} denotes the transversal piezoelectric coefficient of the piezoelectric layer. E_{eff} and ρ_{eff} are the effective Young's modulus and effective mass density of the composite structure, respectively. ϵ is the permittivity in the capacitive gap. g and A denote the gap width and transduction area of MEMS capacitive resonator, respectively. V_{dc} is the dc bias voltage applied to the MEMS resonant structure. T and L represent the thickness and length of the resonator body, respectively. Assumes that the vibrations are perpendicular to the resonator plates.

[†] For gap-variable capacitive MEMS resonators [2], [87].

[‡] For one-port WE mode MEMS resonators [88].

compared to these electrostatic MEMS resonators. Aluminum nitride (AlN) is the most popular thin film piezoelectric layer in the MEMS area [39]–[41], especially for the vibrating MEMS devices. AlN has high acoustic velocity, low temperature coefficient of elasticity (TCE), high thermal conductivity, and good CMOS compatibility. Thus, it offers excellent compromise performance and manufacturability, compared with other piezoelectric materials, such as lead zirconate titanate (PZT) and zinc oxide (ZnO). Table IV shows the detailed comparison between electrostatic and piezoelectric MEMS resonators.

MEMS capacitive resonators have distinguished high Q values, while MEMS piezoelectric, especially piezoelectric on silicon MEMS resonators feature low motional resistance. A low motional impedance resonator can become nonlinear at low voltage levels and thus have poor power handling. Piezoelectric transduction is linear while electrostatic transduction is nonlinear. This allows a piezoelectric resonator to handle much larger powers when compared to an electrostatic resonator with equivalent motional impedance. It is this difference that accounts for the fact that the phase noise of piezoelectric MEMS oscillators is on par with electrostatic MEMS oscillators despite the orders of magnitude lower Q s.

The performance comparison of the MEMS capacitive and piezoelectric resonators reported in the literature is illustrated in Table V. Materials used for the vibrating structure, transduction mechanism, operating mode, resonant frequency, Q , impedance at resonance, as well as the bias voltage and gap width for capacitive resonators are listed in this table. As shown in this table, the capacitive resonators have very high Q , even up to few millions for bulk mode resonators. However, their motional resistances are quite large compared to the piezoelectric resonators. The piezoelectric resonators demonstrate outstanding low motional resistance, while they

TABLE V
PERFORMANCE COMPARISON OF THE REPORTED MEMS RESONATORS PUBLISHED IN THE LITERATURE

Reference	Material	Transduction mechanism	Operating mode	Frequency (Hz)	Bias voltage (V)	Gap width (nm)	Q	R_m (Ohm)
[89]	SCS bulk	Capacitive	Flexural	5.24E+5	2.4	-	5.2E+4	9.00E+4
[90]	Polysilicon	Capacitive	Flexural	17.8E+6	2.5	80	7,200	2.61E+4
[73]	Polysilicon	Capacitive	Transverse	68.5E+6	25	90	1.5E+4	1.34E+4
[75]	SCS bulk	Capacitive	Torsional	1.94E+7	1	130	2.2E+5	1.19E+4
[91]	Polysilicon	Capacitive	Contour-mode	24.4E+6	3	90	6.8E+4	1.32E+4
[92]	SCS bulk	Capacitive	SE	13.1E+6	100	750	1.3E+5	4.47E+3
[93]	SCS bulk	Capacitive	Lamé	5.17E+6	60	2000	3.7E+6	5.00E+4
[87]	SCS bulk	Capacitive	Lamé	6.35E+6	60	2000	1.7E+6	6.14E+4
[14]	SCS bulk	Capacitive	Lamé	4.13E+6	50	2000	1.5E+6	1.31E+4 [†]
[94]	SCS bulk	Capacitive	WE	8.12E+7	10	65	4.0E+4	900
[95]	AlN bulk	Piezoelectric	Lamb wave	1.02E+9	N.A.	N.A.	1,893	49
[96]	AlN bulk	Piezoelectric	Contour-mode	6.90E+8	N.A.	N.A.	3,900	88
[97]	ZnO on SCS bulk	Piezoelectric	Flexural	1.72E+6	N.A.	N.A.	6,200	-
[98]	AlN on SCS bulk	Piezoelectric	Flexural	3.33E+4	N.A.	N.A.	5,000	6.50E+3
[99]	AlN on SCS bulk	Piezoelectric	Flexural	1.54E+6	N.A.	N.A.	2.5E+4	-
[100]	AlN on fused silica	Piezoelectric	Radial breathing	4.88E+6	N.A.	N.A.	2.6E+4	357
[101]	GaN on SCS bulk	Piezoelectric	2 nd -order thickness mode	802.5E+6	N.A.	N.A.	1,850	-
[85]	ZnO on SCS bulk	Piezoelectric	LE	90.4E+6	N.A.	N.A.	3,400	600
[102]	AlN on SCS bulk	Piezoelectric	LE	9.66E+5	N.A.	N.A.	1.4E+5	200
[86]	AlN on SCS bulk	Piezoelectric	Lateral extensional	3.57E+7	N.A.	N.A.	1.0E+4	350
[103]	AlN on SCS bulk	Piezoelectric	Lateral extensional	7.24E+8	N.A.	N.A.	2,000	750
[104]	AlN on SCS bulk	Piezoelectric	Lateral extensional	4.97E+8	N.A.	N.A.	3,800	600
[105]	AlN on SCS bulk	Piezoelectric	Lateral extensional	2.71E+7	N.A.	N.A.	7,500	264
[106]	AlN on SCS bulk	Piezoelectric	Lateral extensional	7.68E+7	N.A.	N.A.	1.1E+4	80
[11]	AlN on SCS bulk	Piezoelectric	WE	2.72E+7	N.A.	N.A.	1.5E+4	32.64

[†] Theoretically calculated value based on the reported data.
N.A. means not applicable.

show medium Q s. Piezoelectric film-on-SCS based resonators combine both the advantages of high Q from capacitive resonators and low impedance from the piezoelectric resonators.

III. FABRICATION AND PACKAGING

Leveraging advances in CMOS technology, various approaches of making MEMS resonators have been demonstrated. Key advantages of MEMS over quartz crystal resonators are related to the way MEMS resonators made. The wafer level, batch fabrication used in MEMS allows for realization of resonators with compact size and low cost. Most of the MEMS resonator fabrication approaches are CMOS-compatible, which is friendly for achieving the monolithic integration of MEMS resonators and their application specific integrated circuits (ASICs). The wafer level process also enables higher levels of system integration, such as MEMS and CMOS combination through wafer-level bonding approaches [107].

The most common transduction types used in MEMS resonators are electrostatic/capacitive and piezoelectric. For electrostatically transduced silicon devices, some of the popular processes reported in literature are surface micromachining [50], [108]–[110], Silicon-on-Insulator (SOI) & Cavity-silicon-on-insulator (CSOI)-based processes [111], [112], and High Aspect-Ratio combined Poly and Single-crystal Silicon (HARPSS) technologies [47], [72], [113], [114]. The SOI & CSOI-based and HARPSS technology are the typical bulk micromachining technologies, which are unique processes in MEMS field. For piezoelectrically

transduced devices, thin-film piezoelectric on silicon (TPoS) platform is a commonly used process to fabricate the piezoelectric MEMS resonators [11], [85], [115], [116].

For MEMS vibrating devices, especially for MEMS resonators, wafer-level vacuum packaging is essential. Vacuum packaging can not only protect the microstructure from mechanical damage to ensure the long-term stability and reliability of the MEMS resonators, but also reduce the air damping for improved performance [14]. There are two primary methods for encapsulating MEMS devices at the wafer scale. One is bonding a cap wafer on top of a separated device wafer. Currently, the most commonly used bonding approaches for MEMS resonators are fusion bonding and Aluminum-Germanium (Al-Ge) eutectic bonding [117]. The other is the deposition of some sort of thin-film encapsulation layer directly on the device wafer. The most popular used thin-film encapsulation method is the Epitaxial seal (Epi-Seal), which has been successful for high-stability timing devices [12], [118]–[120].

A. Surface Micromachining

MEMS resonators fabricated using polysilicon surface micromachining process have been reported in 1980s [121]. In surface micromachining process, the microstructures are built by deposition and etching structural layers over a substrate [122]. Polysilicon is most commonly used structural material while silicon dioxide is used as a sacrificial layer.

The main fabrication steps for a typical polysilicon surface micromachining process are presented in Fig. 3. After the

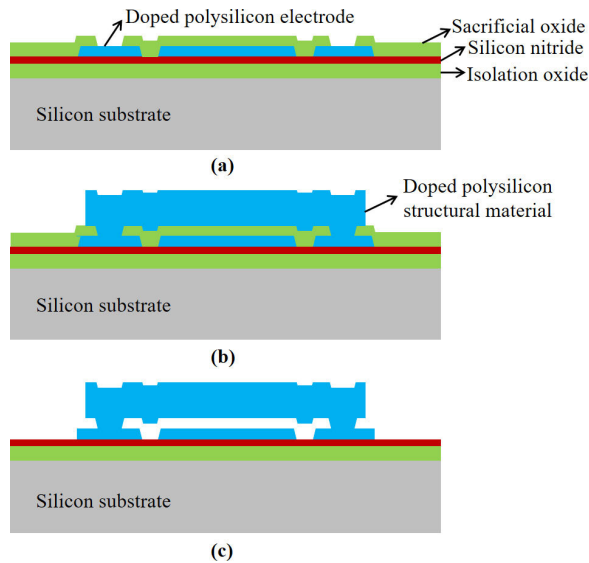


Fig. 3. Cross-sectional views of the process flows of MEMS resonators based on polysilicon surface micromachining technology: (a) Sacrificial oxide deposition and patterning to expose the anchors areas, (b) Doped polysilicon structure layer growth and patterning, (c) Release of polysilicon structure by removal of sacrificial oxide.

isolation oxide and silicon nitride depositions, a doped polysilicon layer is grown and patterned to form the bottom electrodes. Following a sacrificial oxide deposition and patterning to expose the anchor areas, a doped polysilicon structure layer is grown and patterned to make the MEMS resonator structures. Finally, the resonator structures are released by removal of the sacrificial oxide. Narrow gaps can be well defined since the thickness of the sacrificial oxide layer can be easily controlled in the deposition step. The process in Fig. 3 enables out-of-plane sensing for MEMS resonators. The process can also be modified to realize lateral capacitive gaps for the laterally vibrating MEMS resonators, as described in details in [61], [63], [123]. Polysilicon surface micromachining process has been adopted to mass-produce the MEMS resonators in early this century [124], [125].

B. SOI and CSOI Based Process

Compared to the polysilicon surface micromachining, SOI and CSOI based processes are bulk processes, in which single-crystal silicon (SCS) is the structure material and typically use deep reactive ion etching (DRIE) as the main method for defining the capacitive gaps and the resonator structures. SCS is a more attractive structural material for MEMS resonators compared to polysilicon due to its high mechanical Q and low internal stress [47].

A previous version prior to SOI-based process to make the MEMS resonators is the Single-Crystal Reactive Etching And Metallization (SCREAM) process [126]. SCREAM is a single wafer process that is capable of fabricating SCS structures. However, the structure thickness cannot be well controlled in SCREAM process.

SOI technology was originally developed to avoid charge leakage in P-N junctions in the CMOS circuits. While SOI wafers are also attractive to MEMS applications, since they

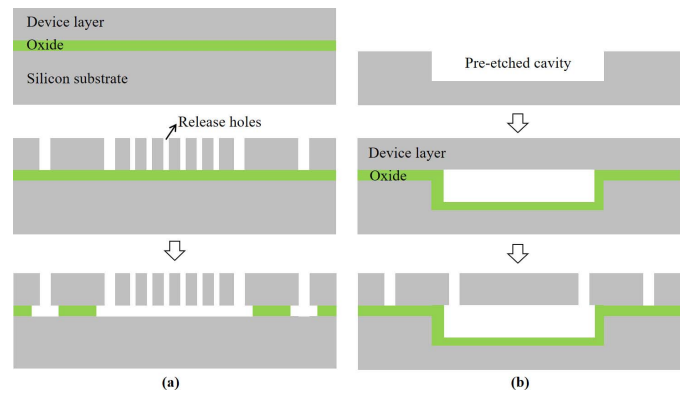


Fig. 4. Cross-sectional views of the process flows of MEMS resonators based on: (a) SOI wafer and (b) CSOI wafer.

provide robust SCS device layer as the structure material for MEMS devices, precise thickness control of structural layers, almost stress free and built-in sacrificial oxide layer. In SOI based process, MEMS resonators are defined using DRIE followed by structure releasing by using hydrofluoric (HF) acid or vapor hydrofluoric (VHF), as shown in Fig. 4(a). Release holes are usually necessary for MEMS devices with large movable structures in SOI based process, due to the limitations on the release areas in the buried oxide etching step. An alternative approach to avoid the release holes could be making a trench underneath the MEMS suspended structures by etching through the substrate from back using DRIE [87], [93].

On the other hand, CSOI wafers have pre-etched cavities which simplify the release of movable structures for MEMS devices [10], [112]. Figure 4(b) illustrates the cross-sectional views of the process flows for MEMS resonators based on a CSOI wafer. Cavities are pre-etched in a silicon substrate and a device silicon wafer is bonded on the substrate using a fusion bonding approach. After grinding and polishing of the device silicon wafer to the required thickness, DRIE is used to pattern the resonator structures. CSOI technology provides the design freedom of MEMS resonators and removes some of the restrictions of using SOI for MEMS devices.

However, the thickness tolerance of the device layer of commonly used SOI and CSOI wafers is quite large, usually between 1% and 10%. A major drawback of SOI and CSOI based process is the frequency variation caused by the thickness tolerance of the device layer of the SOI and CSOI wafers used. While there are new advances, such as Okmetic E-SOI technology, that promise better thickness uniformity of device layer of SOI and CSOI wafers. It can target thick wafers with $\pm 0.1 \mu\text{m}$ uniformity.

C. HARPSS Technology

For capacitive MEMS resonators, thick device layer and narrow gaps are preferred, in order to reduce the motional resistance of the MEMS resonators [87], [127], [128]. Although thick device layers in a variety of thicknesses even up to $100 \mu\text{m}$ are available to make the MEMS resonators using SOI and CSOI-based process, narrow gaps are still difficult to realize due to the aspect ratio limitations in the DRIE process.

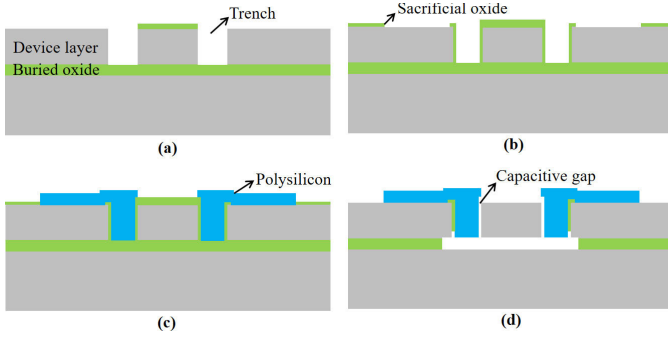


Fig. 5. Cross-sectional views of the process flows using HARPSS-on-SOI approach for fabrication of MEMS resonators. (a) Initial oxide growing and patterning, followed by trench etching using DRIE, (b) Sacrificial oxide thin film growing and patterning, (c) Highly doped polysilicon filling and patterning, and (d) Structure releasing by removing the sacrificial oxide.

HARPSS technology enables the implementation of ultra-high-aspect-ratio (up to few hundreds) capacitive gaps between polysilicon and SCS structures.

In order to realize narrow lateral capacitive gaps in a SOI process, a HARPSS-on-SOI technology is proposed and implemented [72], [114]. SCS is patterned as the MEMS resonator structures and polysilicon is used as the electrodes. The capacitive transduction gaps are defined by using conformal side wall oxide. The fabrication process is shown in Fig. 5. Following initial oxide growth and patterning, trenches are patterned by etching silicon device layer down to buried oxide to define the MEMS resonator structures using DRIE. Then, a sacrificial oxide thin layer is grown and etched back on the surface, whose thickness determines the sizes of capacitive gaps between the resonators body and electrodes. Highly doped polysilicon is deposited to fill the trenches and patterned to form the electrodes. The MEMS resonator structures are finally released and undercut in HF by removing the sacrificial oxide layer and the underlying buried oxide.

D. TPoS Platform

In order to take both of the advantages of high Q of the SCS capacitive resonators and high electromechanical coupling factor of the piezoelectric resonators, thin-film piezoelectric on silicon resonators have been developed [11], [22], [97], [105], [115], [116], [129]–[131]. In TPoS platform, two metal electrodes embedding a thin-film piezoelectric layer inside are stacked on top of the device layer of a SOI or CSOI wafer. Thin-film piezoelectric layers are used as a transducer to excite and sense the acoustic waves in the SCS device layer.

Figure 6 shows the cross-sectional views of the fabrication process for a typical TPoS MEMS resonator. It starts with a CSOI wafer, followed by metal-piezoelectric-metal layers deposited on the device layer of the CSOI wafer. Following top metal patterning and etching to form the top electrode, vias are opened to access the top metal and bottom metal, respectively. Then, an aluminum layer is deposited and patterned to make the electrical contacts. Finally, the MEMS resonator structures are patterned and formed by etching the piezoelectric film-metal-SCS-oxide stacks down the pre-etched cavity in the

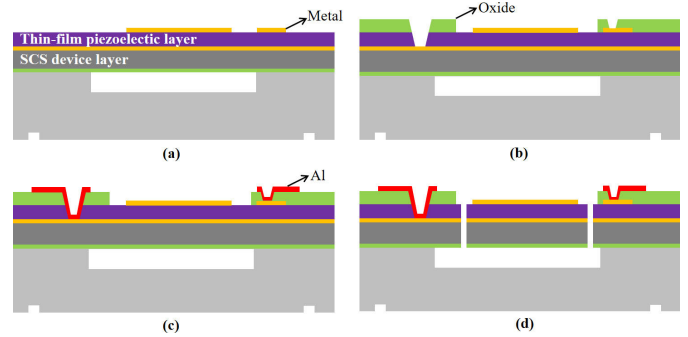


Fig. 6. Cross-sectional views of the process flows using TPoS platform for fabrication of MEMS resonators. (a) Metal-piezoelectric film-metal sandwiched layers deposition on CSOI wafer, followed by top metal patterning to form the top electrode, (b) Vias opening to access the top metal and bottom metal, respectively, (c) Aluminum deposition and patterning to form the electric contacts, and (d) MEMS resonator structure patterning and piezoelectric film-metal-SCS-oxide stacks etching to define and release the active structure.

CSOI wafer. For SOI wafers, backside DRIE could be used to release the MEMS resonators by etching the silicon substrates underneath the resonator body [86].

E. Wafer-Level Vacuum Packaging Using Bonding Approach

Various methods of wafer-level packaging for MEMS resonators have been developed based on the bonding approach. Anodic bonding [132], glass frit bonding [10], [125], [133], polymer bonding [129], [134] silicon-silicon direct (fusion) bonding [135], eutectic bonding [11], [136] have been used to provide wafer-level vacuum sealing for MEMS resonators. Among the existing vacuum packaging technologies, fusion bonding has demonstrated high stability and high vacuum. Al-Ge eutectic bonding approach also features good CMOS-compatibility, high bonding yield and good long term reliability.

Figure 7 shows an example of the process flow for MEMS resonators based on the Al-Ge eutectic wafer-level vacuum packaging approach [11]. Firstly, the TPoS MEMS resonators are fabricated starting from CSOI wafer, followed by piezoelectric stacks deposition, vias opening for electrical contact, and MEMS resonator body patterning. Following deep trenches etching using DRIE in a cap wafer, an oxide insulation layer is grown over the trenches, and highly doped polysilicon is deposited on the cap wafer to refill the trenches. After a thick oxide layer is grown and patterned to form the spaces, a Ge layer is deposited and patterned as the redistribution lines (RDL) and sealing ring on the cap wafer. Secondly, the MEMS wafer with fabricated MEMS resonators and the cap wafer with polysilicon trenches are bonded together in a vacuum chamber using the Al-Ge eutectic bonding approach. Following the bonding step, the bonded wafer is grinded on the front side to expose the polysilicon trenches in the cap wafer. Thirdly, after an oxide layer deposition and contact holes opening, an Al layer is deposited and patterned to form the bonding pads.

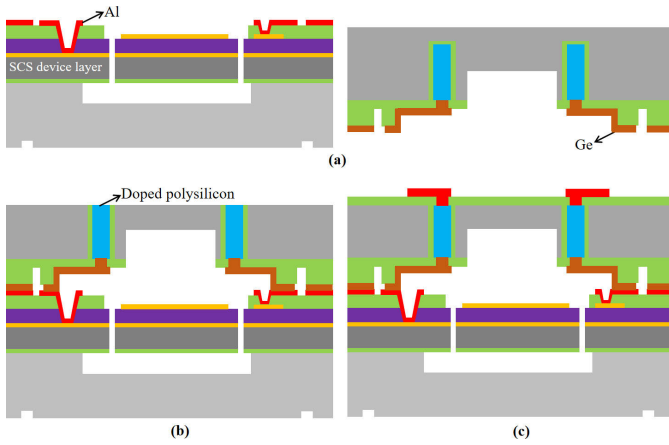


Fig. 7. Cross-sectional views of the Al-Ge wafer-level vacuum packaging-based process flows for fabrication of MEMS resonators. (a) MEMS wafer and cap wafer preparation, (b) Wafer level vacuum bonding using Al-Ge eutectic approach, followed by cap wafer grinding to expose the polysilicon trenches, (c) Following oxide growing and contact opening, aluminum is deposited and etched to form the bonding pads.

F. Wafer-Level Vacuum Packaging Using Epi-Seal Process

Thin-film encapsulations are based on surface micromachining techniques, where a layer is deposited on the device wafer to provide encapsulation. The most commonly used thin-film encapsulation layers are oxide [137], [138], silicon nitride [139], aluminum nitride [140], aluminum, amorphous silicon, and polysilicon [94]. Low pressure chemical vapor deposition (LPCVD) [137], plasma enhanced chemical vapor deposition (PECVD) [94], sputtering [139], [140], and epitaxial processes [12], [118]–[120] are usually used to deposit these thin-films.

Most of the thin-film encapsulations are conducted at a low temperature usually below 500 °C. There are volatile organic and water residues in the package for packaging processes under low temperature environment, which leads to mass loading or degassing issues and hence frequency shift of MEMS resonator, degrading the long-term stability of the devices. In order to achieve a high stability and robust thin-film encapsulation, an Epi-Seal platform has been developed by Stanford University and Bosch Palo Alto Research and Technology Center in the early 2000s, and is now in use at SiTime [118]–[120].

Epi-Seal process starts with a SOI wafer, and trench etching by DRIE is followed to define MEMS resonator structures. Next, an oxide layer is filled over the trench and patterned on the surface for electrical contact to the device layer. Following a thin polysilicon layer deposition over the oxide using epitaxial, vent holes are etched into the polysilicon to access the resonator cavity. Then, oxide is removed through the vent holes by using VHF and the movable structures are released by undercutting the buried oxide underneath the resonator body. A second thick epitaxial polysilicon layer is deposited over the vent holes to seal the device, followed by polysilicon etching to define the isolation trenches for electrical contacts. Finally, aluminum electrical contact pads are deposited and patterned. To further reduce the pressure in the sealed cavity,

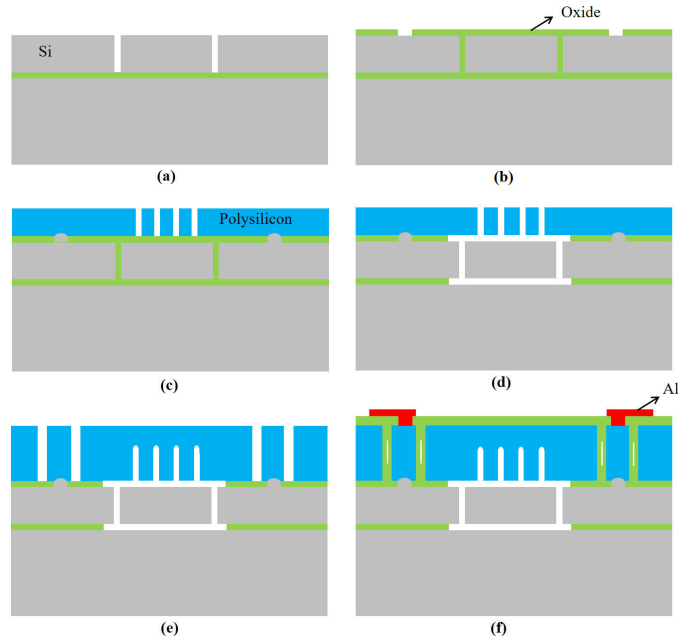


Fig. 8. Cross-sectional views of the Epi-Seal process flow for fabricating MEMS resonators. (a) Trench etching using DRIE to define the MEMS resonator structures, (b) Oxide growing over the trench, followed by oxide patterning for electrical contact to the device layer, (c) Epitaxial polysilicon deposition over the oxide, and vent holes etching into the polysilicon, (d) Oxide release through the vent holes, (e) Thick epitaxial polysilicon deposition over the vent holes, and polysilicon etching to define the isolation trenches for electrical contacts, (f) Oxide growth and patterning, followed by aluminum deposition and etching to form the bonding pads.

the remaining H_2 is diffused out by annealing in a nitrogen flow at around 400 °C. Figure 8 shows the cross-sectional views of the Epi-Seal process flows for fabrication of MEMS resonators.

The epitaxial polysilicon is deposited at a high temperature above 1000 °C in a clean vacuum environment, which leads to a low cavity pressure (sub-Pa) and native oxide free on the silicon surface. In the Epi-Seal platform, only SCS, polysilicon and oxide are used and all of them are grown and annealed at high temperature, which avoids the material instability issue. Above all, MEMS resonators fabricated using the Epi-seal process, have demonstrated good long term stability with no noticeable long term drift [12], [119], [120].

IV. KEY TECHNOLOGIES FOR HIGH PERFORMANCE RESONATORS

This section describes the details of the key technologies to improve the major performance metrics of the MEMS resonators. The common energy dissipation mechanisms of the MEMS resonators are discussed and various technologies to improve Q_s of MEMS resonators are reviewed. Approaches to reduce the R_m of MEMS resonators are compared and methods for improving the temperature stability are reported in details. Various technologies for frequency trimming are illustrated in order to reduce the intimal frequency variations and improve the uniformity. Above all, key technologies for making MEMS resonators with high performance are reviewed and compared in this section.

A. Quality Factor

Q is one of the most important characteristics for MEMS resonators. Generally, a higher Q is desired for most applications, especially for the timing and sensing applications.

According to Leeson's equation [1], [141], [142], the single side band phase noise, $L(\Delta f)$, in an oscillator is expressed as:

$$L(\Delta f) = \left(\frac{2Fk_bT}{P_0} + \frac{b_1}{\Delta f} \right) \left(1 + \frac{f_c^2}{4Q_{loaded}^2 \Delta f^2} \right) \quad (6)$$

where f_c is the oscillator carrier frequency, Δf is the specific offset frequency from f_c , F is noise figure of the amplifier, k_bT is the thermal noise, Q_{loaded} is the loaded Q of the resonator, b_1 is an empirical parameter, and P_0 is the signal power at the input of amplifier.

The phase noise, a key indicator of the short-term stability of an oscillator, is inversely proportional to Q_{loaded} and the power delivered to the sustaining amplifier. In order to improve the MEMS oscillator's phase noise performance, the frequency-selective MEMS resonators are required to have high Q_{loaded} and large power handling capability. High Q_{loaded} is imperative as it enables low close to carrier frequency noise. Thus, MEMS resonators are required to have high unloaded Q , which typically results in high Q_{loaded} are therefore beneficial in attaining low phase-noise.

For MEMS resonators, Q is strongly affected by several energy dissipation mechanisms, such as anchor loss, thermo-elastic dissipation, air damping, and intrinsic material loss [14], [87], [91], [143]–[148]. The overall Q is determined by the reciprocal sum of all energy losses from each energy dissipation mechanism, which is usually expressed as:

$$\frac{1}{Q} = \frac{1}{Q_{anc}} + \frac{1}{Q_{TED}} + \frac{1}{Q_{air}} + \frac{1}{Q_{mat}} \quad (7)$$

1) *Anchor Loss*: When MEMS resonator is vibrating, not all mechanical vibration energy is reflected back to the MEMS resonator body. Shear forces and bending moments on the anchors will stimulate elastic waves propagating into the substrate. These mechanical waves are absorbed by the substrate, resulting in energy dissipation [149]–[151]. This kind of energy dissipation in MEMS resonator is defined as anchor loss or support loss.

Various methods have been reported to reduce the anchor losses in MEMS resonators. The first step is to minimize the mechanical vibration propagating into the substrate, such as anchoring the resonator with tethers at its nodal or quasi-nodal points [123], [152]–[154]. A further step is to reflect back the elastic waves propagating at the anchor-substrate boundary as much as possible. The fundamental principle is based on the acoustic impedance mismatch between the resonator and the tethers [155], [156]. Quarter wavelength tether [50], [148], [152], material-mismatched support [157], acoustic reflection cavity [151], [155], [158] and phonon crystal (PnC) tethers [95], [159]–[163] are commonly used approaches in this step.

Placing the anchors at the nodal points is the most common approach to minimize the anchor losses for MEMS resonators. The nodal points have minimal displacement, thus less

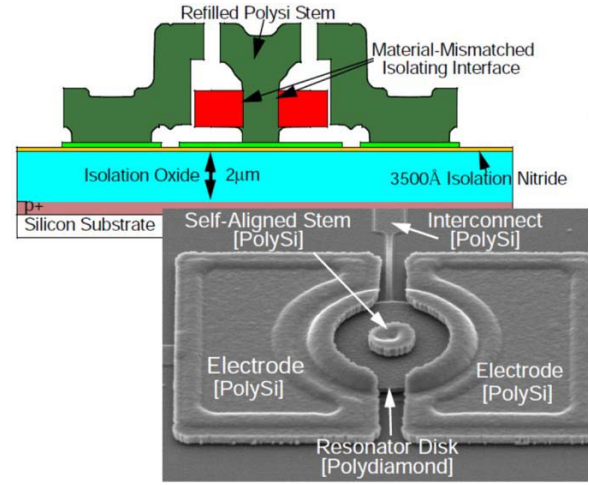


Fig. 9. Cross-sectional view and SEM image of a diamond MEMS disk resonator with material-mismatched isolating support for reducing anchor losses. Reprinted with permission from [157].

mechanical waves will propagate into the substrate, resulting in low anchor losses. However, this method may not apply to certain resonators, in which the nodal points are not around the resonator edges, causing difficulty in tether designs.

Quarter wavelength tether is an effective approach to minimize the energy loss through these tethers. Similar to the theory used for transmission lines, quarter wavelength tether is assumed as the best acoustic isolation, since the complete in phase reflection occurs as the tether length equals to a quarter acoustic wavelength, or $\lambda/4$. Therefore, there is hardly any energy dissipation to the substrate through the tethers. The designs of quarter-wavelength supports attached at the nodal points for free-free beam MEMS resonator [50] and hollow-disk ring resonator [152] have been demonstrated to adeptly reduce the anchor dissipations. For specific resonators, the tether dimensions can be optimized in order to provide improved Q_{anc} [148]. However, quarter-wavelength design results in extremely long tether structures, usually in tens to hundreds of micrometers, which is counter to minimization and leads to a decrease in the mechanical stability of the devices.

Material-mismatched support is also used to minimize the anchor losses based on the acoustic impedance mismatch between the resonator structure and anchor [157]. The resonator structure and anchoring stem are made with different materials. The acoustic impedance mismatch between these two suppresses the energy from the resonator to the stem, thus reducing anchor losses and allowing high Q . Figure 9 shows a MEMS disk resonator with material-mismatched support for achieving high Q [157]. The disk resonator structure is made of polydiamond while the supporting stem is made of polysilicon. The acoustic velocity of polydiamond is $2.19 \times$ higher than that of polysilicon, which leads to a large acoustic impedance mismatch and allows high Q disk resonator.

Acoustic reflection cavity is another alternative approach to reduce anchor losses, which could be achieved by etching

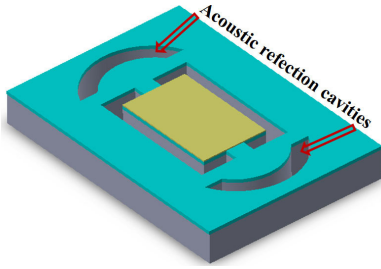


Fig. 10. Schematic illustration of a MEMS resonator with in-plane acoustic reflection cavities for anchor losses reduction.

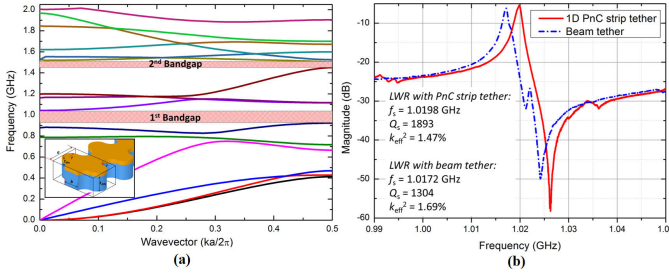


Fig. 11. AlN MEMS resonator with one dimensional phononic crystal tethers. (a) Simulated dispersion curves and complete band gaps for the one dimensional gourd-shape PnC structure. The inset is the schematic view of the gourd-shape PnC structure. (b) Measured transmission curves of the MEMS resonators with PnC tethers and beam tethers. Reprinted with permission from [95].

trenches into the substrate [151], [158]. The basic mechanism is to reflect back a portion of the elastic waves at the anchor boundary due to the discontinuity in the acoustic impedance caused by the acoustic cavity, as shown in Fig. 10. Analogous to the quarter wavelength tether approach, an acoustic reflection cavity suppresses acoustic waves to propagate into substrate and confines energy dissipation into substrate. A lateral-extensional mode MEMS resonator with in-plane acoustic reflection cavity for reducing anchor losses has been reported in [158]. The average measured Q demonstrates an enhancement of over 560% for a 110 MHz MEMS resonator with optimized acoustic reflection cavities.

Anchor losses can also be reduced by anchoring the MEMS resonators using PnC tethers [95], [159]–[163]. PnC tethers is a promising way to restrain the acoustic wave propagation in the supporting tethers, since they can arouse complete band gaps in which the transmissions of the elastic waves are prohibited. Thus, the vibration energy is retained in the resonator body, reducing the anchor losses into the substrate. Figure 11 shows an AlN Lamb wave MEMS resonator with one dimensional gourd-shape PnC crystal tethers [95]. Compared to the MEMS resonator with straight beam tethers, the measured Q shows an improvement of over 45% for the resonator using the PnC crystal tethers. However, the critical dimension in the PnC structures is usually in the range of hundreds of nanometers, especially in the sub-GHz or above GHz frequency range, which brings challenges in the fabrication process.

SMR is built on top of quarter-wave thick stacked layers of alternating low and high acoustic impedance materials [42], [164]. The stacked layers are referred to as the

Bragg reflector stack, providing acoustic isolation from the substrate [41], [155]. The Bragg reflector stack is similar to a one-dimensional PnC crystal structure in the vertical direction.

Anchor losses are largely dependent on geometry, resonant mode shape, and way of packaging [150]. Anchor losses have less contribution to the overall energy losses in low frequency MEMS resonators with proper anchor design [147]. While anchor losses are believed as the dominant energy loss mechanism for high frequency MEMS resonators [148]. Anchor losses are expected to be relatively temperature independent [147], since the propagation of elastic waves into the anchor is dependent on the anchor dimensions, which does not change much with temperature.

2) *Thermoelastic Damping*: Thermoelastic damping (TED) results from the thermal gradients created in a structure due to nonuniform strain fields when the resonator is vibrating [150], [165]–[171]. Temperature in the compressed regions increases, and that of the stretched regions decreases, resulting in temperature gradients in the resonator structure. Temperature gradients lead to thermal heat flow between the hot and cold regions in the resonator and energy dissipation occurs. Therefore, the basic mechanism of TED is the coupling of the thermal and mechanical modes of a resonator when it is vibrating. It is worth to note that although TED occurs in many flexural and longitudinal mode resonators, some resonators operating in pure shear and torsion, such as the wineglass mode, do not experience TED [37].

For a simple clamped-clamped beam vibrating in its fundamental flexural mode, the resulting Q_{TED} has been firstly evaluated by Zener in 1930's [165], [166], expressed as:

$$Q_{TED} = \frac{c_v}{E\alpha^2 T_0} \frac{1 + \omega_m^2 \tau_{th}^2}{\omega_m \tau_{th}} = Q_{TED,mat} Q_{TED,des} \quad (8)$$

with

$$Q_{TED,mat} = \frac{c_v}{E\alpha^2 T_0}, \quad Q_{TED,des} = \frac{1 + \omega_m^2 \tau_{th}^2}{\omega_m \tau_{th}} \quad (9)$$

where c_v is the heat capacity, E is the effective Young's modulus, and α is the coefficient of thermal expansion of the resonator material, respectively. T_0 is the average temperature, ω_m is the mechanical angular resonant frequency and τ_{th} is the thermal time constant of the material. An explicit expression of TED for flexural mode resonator with one-dimensional model has been reported in [167]. A fully coupled thermomechanical model in both two and three dimensions in a solid has been presented in [169].

According to Eq. (8) and Eq. (9), Q_{TED} can be divided into two parts. One is the material related term $Q_{TED,mat}$, and the other is the design related term $Q_{TED,des}$. $Q_{TED,mat}$ is independent of the resonator dimension and is dependent only on material parameters. While $Q_{TED,des}$ is dependent on the resonator design. $Q_{TED,des}$ as well as Q_{TED} has a minimum at $\omega_m = \frac{1}{\tau_{th}}$, where maximum thermal energy will be dissipated as thermal flows during every cycle due to the increase of the thermal-mechanical coupling at this frequency. Therefore, a simple way to minimize the TED is to keep ω_m away from $\frac{1}{\tau_{th}}$ as far as possible in the design stage.

Geometry design with slots engineered at specific locations within the resonator structure has been demonstrated as an

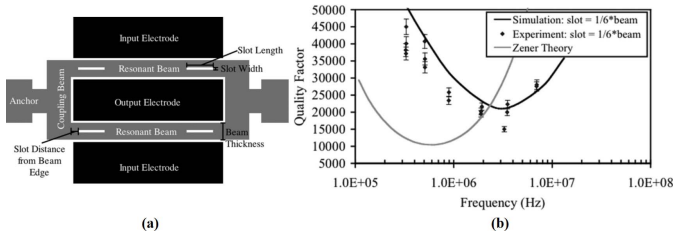


Fig. 12. TED reduction based on geometry design using slots into the resonator structure. (a) Schematic top view of a clamped-clamped flexural mode resonator with two slots in the resonant beams, and (b) Measurement results of resonator with slots in beams, compared to finite element simulation and simple Zener theory. Reprinted with permission from [171].

effective method to reduce TED [171]. Figure 12(a) shows a clamped-clamped flexural mode MEMS resonator with two slots in the resonant beams for TED reduction. The designed slots disrupt the thermal flow across the beam and thus altering the energy dissipations [170], [171]. The slots located at the regions of high strain can greatly alter the coupling of thermal and mechanical eigenmodes. Compared to the resonators without slots, the measured Q s show $\sim 4\times$ enhancement for the resonators with additional slots in proper locations, as shown in Fig. 12(b).

For piezoelectric MEMS resonators, TED is more complicated than silicon capacitive resonators, since more layers and multiple materials are involved [172], [173]. The metal material, electrode coverage ratio as well as metal thickness has large effect on resonator's TED [172]. The impact of the metal electrodes on Q of AlN piezoelectric resonators has been investigated. The results show that lower electrode coverage ratio and smaller electrode thickness are corresponding to higher Q_{TED} .

TED is not only dependent on the mechanical resonant frequency and geometry, but also dependent on the operating temperature. This is due to the fact that TED is also a material related energy loss mechanism. The material properties, such as α , c_v and E , change extremely rapidly throughout the normal operation temperature range (-40°C to $+125^\circ\text{C}$) [144]. Thus, Q_{TED} is largely temperature dependent and varies inversely with temperature with a coefficient in the range of 2-3.

An interesting work has been reported that TED of silicon MEMS resonators can be disabled or eliminated at certain temperatures, where the coefficient of thermal expansion of silicon equals zero [150], [168]. Thereby, the coupling between the thermal and mechanical modes are disabled, thus no TED occurs. When operating at these temperature, usually at -150°C for SCS, the MEMS resonators show no TED. It provides a possible way to improve the overall Q of silicon resonators by eliminating the resonator's Q_{TED} through controlling the operating temperature, although it may not be quite practical for many applications.

3) *Air Damping*: Air damping has been verified as one of major energy loss mechanisms for MEMS resonators operating in atmosphere. The basic mechanisms of air damping include squeeze film damping and slide film damping [174], [175]. Usually, squeeze film damping is especially

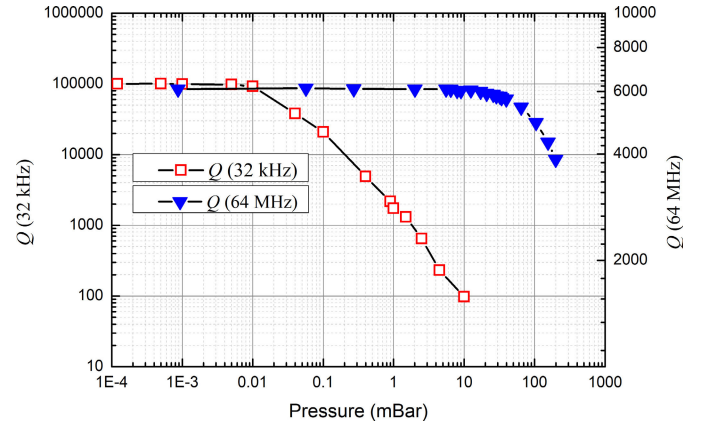


Fig. 13. Comparison of measured Q versus operating pressure curves for a low and a high frequency MEMS resonator.

significant in the transduction gaps of gap-variable capacitive MEMS resonators, and slide film damping is the dominant air damping mechanism in overlap-variable capacitive or piezoelectric MEMS resonators.

For MEMS resonators, Q_{air} is commonly inversely proportional to the operating pressure and proportional to the resonant frequency [144], [150], [174]. MEMS resonators vibrating in low frequency are more sensitive to the air damping compared to the high frequency MEMS resonators. Figure 13 shows the dependence of Q s on operating pressures for a 32 kHz and a 64 MHz MEMS resonator. For the 32 kHz low frequency resonator, the Q starts to degrade at around 0.01 mbar. While Q of the 64 MHz high frequency resonator is almost constant up to a pressure level of 10 mbar. Below these critical pressures, other energy loss sources, such as anchor loss or TED, dominate the overall energy dissipation of MEMS resonators. Measured Q s for both low and high frequency resonators drop slowly above the critical pressure and then fast with the pressure, indicating that the air damping becomes dominant loss mechanism in the pressure range above this pressure. Therefore, a higher vacuum level is usually required, especially for the low frequency resonators. Q_{air} is further reduced as the surface to volume ratios become significantly larger [176]. The reason is that more energy is dissipated by air damping with increased surfaces.

In most cases, vacuum packaging is employed to eliminate the air damping and thus improve the resonator's Q . Chip level and wafer level chip scale vacuum packaging are the most commonly used approaches for vacuum sealing of MEMS resonators [1], [175]. Wafer-level vacuum packaging provides a competitive and outstanding feature for MEMS resonators, compared to the chip-level packaging approach used in the quartz crystal resonators.

4) *Intrinsic Material Loss*: The $f \times Q$ of a given MEMS resonator is limited by the fundamental intrinsic material dissipations, which consists of phonon-electron and phonon-phonon interactions [36], [145], [177], [178]. For silicon or piezoelectric based MEMS resonators, dissipation due to phonon-phonon interactions (Q_{p-p}) is of most significance intrinsic material loss mechanism. Phonon-phonon interactions determine the upper limit of $f \times Q$ in MEMS resonators.

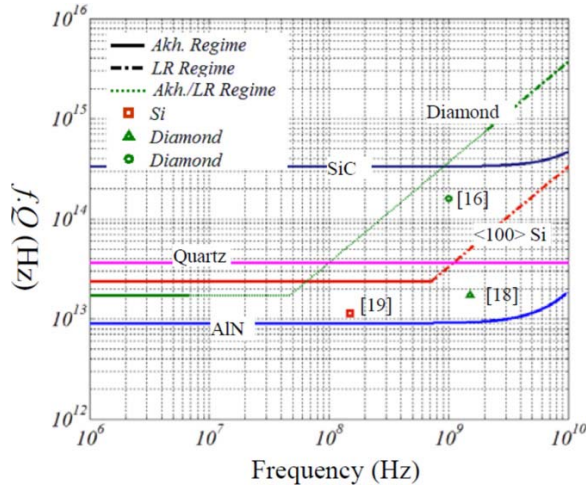


Fig. 14. Comparison of $f \times Q$ versus frequency curves for silicon, quartz, and diamond. Reprinted with permission from [145].

Phonon-phonon interactions can be divided into two main regimes, namely Akhiezer and Landau-Rumer regimes [36], [145]. In Akhiezer regime, the acoustic wave phonon interacts with the ensemble of thermally excited phonon in the material. For longitudinal waves in this regime, $f \times Q$ is a constant set by the resonator material. For SCS, the upper limit of $f \times Q$ has been calculated to be 3.9×10^{13} Hz [36]. In Landau-Rumer regime, the acoustic phonon interacts with individual thermal phonon. $f \times Q$ in Landau-Rumer regime is linearly increased with operating frequency. The transition from Akhiezer to Landau-Rumer regime occurs at high frequency, usually at around 1 GHz for most MEMS materials as shown in Fig. 14.

B. Motional Impedance

Motional impedance is one of important remaining issues to render the development of MEMS resonators in RF front ends. In the wireless communication applications, matching impedances in the ranges of 50 Ohm and 75 Ohm are often required. Motional impedance of present high frequency MEMS resonators, especially for capacitive MEMS resonators, are usually in the range of hundreds to thousands Ohms, which prevents them from directly coupling to the antennas.

As described in Eq. (6), the close-to-carrier frequency as well as far-from-carrier frequency phase noise can be reduced by increasing the input signal power P_0 of the amplifier. One way is to increase the driving power level of the resonator to increase the input power, which requires the resonators to have high power handling capability [1], [179]. Another effective way is minimizing the power attenuation of the resonator, which can be realized by reducing the impedance ratio between R_m and impedances that are parallel to R_m . Thus, for a given resonator, low R_m is important to minimize the phase noise of the oscillator.

Moreover, low R_m allows for low amplifier gain needed in the sustaining amplifier, resulting in less power consumption. Therefore, a low R_m is also preferred to minimize the power consumption in the oscillator loop. Various methods and

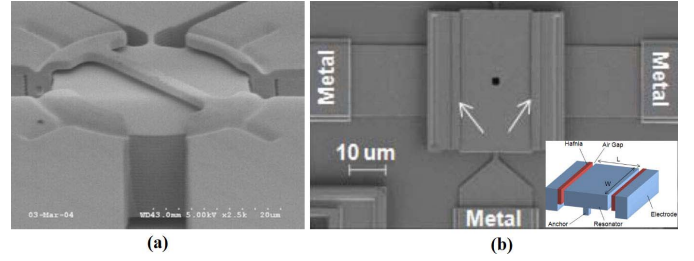


Fig. 15. Motional impedance reduction by using narrow capacitive transduction gaps. (a) SEM image of an 18 μm thick SCS bulk mode disk resonators with capacitive gap width of 160 nm. Reprinted with permission from [114]. (b) SEM image of a bulk mode resonator, indicating the 10 nm air and 90 nm hafnia capacitive gap. Reprinted with permission from [128]. The inset is the cross-sectional view of the resonator structure after release.

approaches have been implemented to reduce the R_m of the MEMS resonators. Increasing the electromechanical coupling factor is one of most often used methods to reduce the R_m of the MEMS resonators.

1) *Manipulation of the Electromechanical Coupling Factor:* According to Eq. (5), R_m of the MEMS resonator is dependent on the mechanical properties as well as the electromechanical coupling factor, η , of the resonator. One effective way to reduce R_m is to enhance η , since R_m is inversely proportional to the square of η . According to Table IV, R_m of a capacitive resonator is inversely proportional to the square of the bias voltage V_{dc} and transduction area A , and is also dependent on the capacitive gap width d to the fourth power. Narrow gaps, larger transduction area, and high bias voltage are usually used to improve the η , and hence to minimize R_m for the capacitive MEMS resonators [1], [114], [128], [180]–[182].

Increasing the bias voltage is a simple and direct way to reduce R_m . While charge pumps are usually needed to generate high dc voltage, which increases the complexity of the circuit designs. Large transduction areas can be easily obtained by using thick device layers. Using narrow gaps is the most efficient way to reduce the R_m for capacitive MEMS resonators.

Narrow gaps and thick device layers can be achieved based on the HARPSS-On-SOI platform [114], as shown in Fig. 15(a). Reducing gaps and increasing thickness for the capacitive MEMS resonators have been demonstrated. An 18 μm thick SCS bulk mode disk resonator shows over $20\times$ lower R_m compared to a 3 μm thick resonator [114]. The transduction gaps could also be reduced by inserting dielectric layers as reported in [128], [183]. Nanometer gaps have been achieved by dual depositing of 10 nm sacrificial oxide and 90 nm hafnia via atomic layer deposition (ALD) followed by oxide release [128], as shown in Fig. 15(b).

Adopting transduction mechanisms with high η , such as piezoelectric, is also used to achieve a low R_m [1], [11], [88], [184]. This is due to the fact that the coupling factor for a piezoelectric resonator is much higher than the one for a capacitive resonator. Typically, the piezoelectric coupling factor is more than three orders of magnitude larger than the capacitive one [1]. Therefore, piezoelectric resonators tend to have much lower R_m compared to the capacitive resonators.

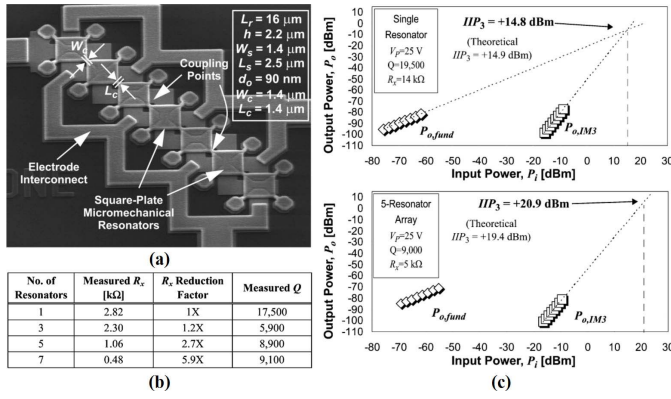


Fig. 16. Mechanically corner-coupled MEMS resonator array for reduced series motional resistance. (a) SEM image of a 72-MHz mechanically coupled array of seven corner-supported square plate resonators. (b) Performance comparison of the MEMS resonator arrays, and (c) Measured output power versus input power curves for the single corner-supported MEMS resonator (top), and an array of five of MEMS resonators (bottom). Reprinted with permission from [73].

The R_m of contour-mode piezoelectric AlN MEMS resonators are measured in the range of 50 Ohm to 700 Ohm [88], comparable to the classical quartz crystal resonators.

2) *Array Design*: Array design is another helpful approach to reduce R_m for both capacitive and piezoelectric MEMS resonators [73], [185]–[187]. Mechanically coupling several resonators into an array composite resonator is an efficient solution to reduce the motional impedance and enhance the power handling capability of the MEMS resonators.

Substantial reduction in R_m has been demonstrated by mechanically coupling a parallel array of corner-coupled polysilicon square plate resonators in [73]. The resonator array consists of seven identical MEMS resonators, mechanically coupled at corners with identical coupling stubs, as shown in Fig. 16(a). Figure 16(b) provides a comparison of the measured results for the single and arrayed resonators. It clearly shows that R_m is decreased with an increase in the number of resonators in the array. The reduction factor is roughly equivalent to the number of the resonators used. Array design can also enhance the linearity and hence power handling capability of the MEMS resonators, as shown in Fig. 16(c).

C. Temperature Stability

In the case of timing applications, temperature stability is extremely important. Most MEMS resonators materials exhibit a large temperature coefficient of elasticity (TCE) and coefficient of thermal expansion, α , which leads to significant temperature-induced frequency variations even with proper packaging. The first-order temperature sensitivity of f_0 can be defined and related to the TCE and α of the resonator's material, which is expressed as:

$$TCF = \frac{1}{f_0} \frac{\partial f_0}{\partial T} \approx \frac{1}{2}(TCE + \alpha) \quad (10)$$

where T is the operating temperature. Typically, the native silicon TCE of ~ -60 ppm/°C (dependent on the resonator design), and α of ~ 2.6 ppm/°C yield a TCF of around

-30 ppm/°C for silicon resonators. In the full industrial range of -40°C to $+85^\circ\text{C}$, the resonant frequency of a silicon resonator could shift by over 3500 ppm, which is not acceptable in the real applications. Therefore, temperature compensation is needed and necessary to minimize the TCF of the MEMS resonators. The most commonly used temperature compensation techniques are based on either passive or active compensation approaches.

In passive compensation approaches, the intrinsic property of the resonating structure is designed, such that the frequency sensitivity due to temperature change is reduced. Passive approaches have gained much of attention, since they do not consume additional power and not need dedicated circuits for the frequency tuning. Composite structure [90], [99], [188]–[194], doping [82], [195]–[201], geometry engineering [202]–[204] are the most commonly used passive compensation methods. Moreover, the TCF of silicon MEMS resonators can also be tuned according to the resonance modes and resonator orientations, since the anisotropic material properties of SCS. The temperature dependences of the elastic constants of doped silicon are extracted by using a suite of different types and orientations of resonators with various doping concentrations [200], [201]. Design rules for TCF tuning or adjustment of similarly doped silicon resonators have been provided.

1) *Composite Structure*: The frequency-temperature sensitivity in silicon MEMS resonators is primarily caused by the large TCE of the material used. Most of the MEMS resonator materials exhibit negative valued TCE . The materials soften, such that the stiffness, and thus frequency of the MEMS resonators decreases with increasing temperature. Leveraging the positive valued TCE of some material, such as silicon dioxide (SiO_2), temperature-insensitive MEMS resonators can be achieved by the addition of SiO_2 in the resonator body to create a composite structure.

One of the earliest demonstrations of temperature compensation using the composite structures dates back to the early 1980s [188]. Such approach was also implemented for FBARs by introducing SiO_2 layer underneath the Al-ZnO-Al piezoelectric stack layers [189]. The first order TCF varies from -1.6 ppm/°C to -0.4 ppm/°C over a temperature range of 40°C to 105°C .

Silicon- SiO_2 MEMS resonators were demonstrated, in which the resonator body was coated with a thermal oxide layer [191], [192]. A composite DETF resonator exhibited a quadratic temperature dependence, with frequency variations of less than 200 ppm, in the temperature range from -55°C to 125°C , as shown in Fig. 17. The frequency-temperature dependence of the composite DETF resonator is comparable to watch-grade quartz-crystal resonators.

Although SiO_2 coating is an effective way for passive temperature compensation of MEMS resonators, the dielectric charge due to the coated SiO_2 could result in significant frequency drifts over time [192], [205]. To avoid the charging-related issues using oxide as the compensation material, a modified approach by burying the oxide layers inside the resonator structure is demonstrated [90], as shown in Fig. 18. The resonator structure consists of highly-conductive polysilicon.

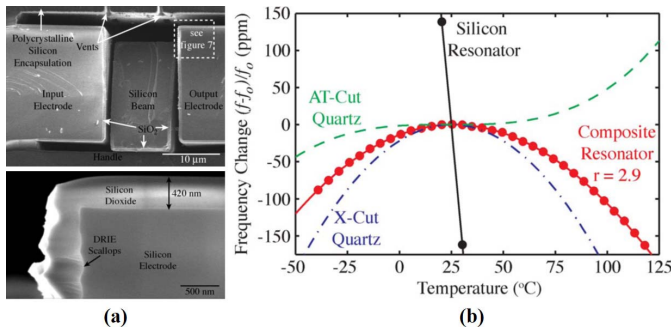


Fig. 17. Temperature-insensitive composite MEMS resonators. (a) SEM images of composite resonator with thermal oxide coating on the resonator body and zoomed-in view of a corner of the electrode, (b) Frequency-temperature measurements of composite DETF resonator with a flexural ratio of $r=2.9$, compared to a silicon resonator, and the response of two common quartz-crystal resonators. Reprinted with permission from [192].

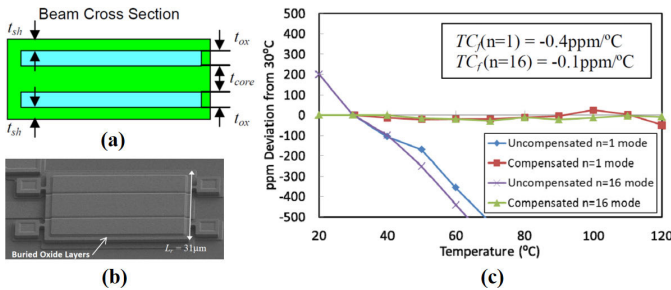


Fig. 18. Temperature compensated MEMS resonator with buried oxide layers. (a) Cross-sectional view of the MEMS resonator with layer thicknesses labeled, (b) SEM image of a fabricated fundamental mode MEMS resonator with buried temperature compensation layers, and (c) Comparisons of the measured frequency versus temperature curves for the compensated and uncompensated MEMS resonators. Reprinted with permission from [90].

Therefore, the design avoids the issues with oxide charging. With the buried oxide layers, the resulting TCF shows a $40\times$ decrease for the compensated fundamental mode resonator, and a $150\times$ reduction for the overtone design, respectively.

For flexural mode resonators, growing or depositing a thin SiO_2 layer over the resonator body is efficient for temperature compensation. However, the thin SiO_2 layer becomes less efficient for temperature compensation in bulk mode MEMS resonators, since the SiO_2 has much smaller elastic modulus compared with SCS or AlN . A bulk oxide compensation approach by embedding the SiO_2 pillars in bulk silicon resonant structure was proposed and reported in [193], [206], [207]. Figure 19(a) and (b) show the scanning electron microscope (SEM) images of a temperature compensated AlN -on- SiO_x resonator. As shown in Fig. 19(c), the compensated AlN -on- SiO_x resonator exhibits an overall frequency variation of 83 ppm over the entire temperature range of -40°C to 80°C , which corresponds to a $40\times$ improvement in the overall frequency variation compared with a non-compensated silicon resonator. By tuning the volumetric ratio of SiO_2 in SiO_x unit, the turnover temperature could also be tailored.

2) *Doping*: The TCF of silicon MEMS resonators is largely governed by the TCE of the resonating silicon

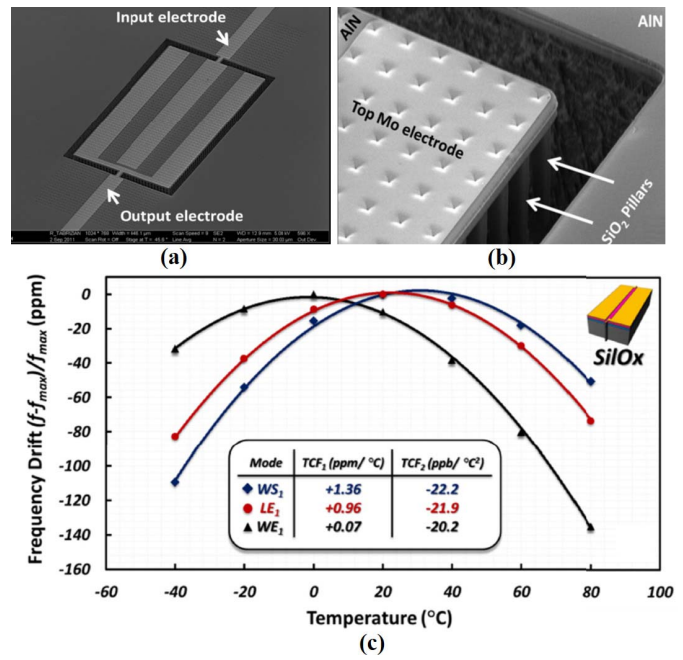


Fig. 19. Temperature compensated MEMS resonator with oxide pillars embedded in resonator body. (a) SEM images of the temperature compensated AlN -on- SiO_x resonator and (b) the zoom-in view of the SiO_2 pillars extruding from the resonator sidewall, (c) Measured frequency-temperature dependence of the three different resonance modes of a SiO_x resonator. Reprinted with permission from [193].

structure, as described in Eq.(10). TCE of silicon is dependent on the charge carrier concentration, due to free charge carrier effects on the elastic constants of silicon [208]. Heavy doping with either p-type or n-type dopants has been demonstrated to reduce the TCF of silicon MEMS resonators. This TCF reduction mechanism is attributed to that heavy doping strains the crystal lattice and shifts the electronic energy bands, resulting in a flow of charge carriers to minimize the free energy, thereby changing the elastic properties [200], [208].

Heavy doping TCF reduction technique was first reported for MEMS silicon bulk acoustic resonators (SiBAR) in 2009 [195], [196]. The TCF of SiBAR was reduced from $-29 \text{ ppm}/^{\circ}\text{C}$ to $-1.5 \text{ ppm}/^{\circ}\text{C}$ on $5 \mu\text{m}$ thick devices using degenerate boron doping, with a resistivity of around $10^{-4} \text{ Ohm}\cdot\text{cm}$, equivalent to a boron density of $\sim 2 \times 10^{20} \text{ atoms}/\text{cm}^3$, as shown in Fig. 20(a). In order to avoid the thickness limitation and long annealing time issue in the degenerate boron doping process, a boron-assisted Al thermomigration is proposed as a faster alternative to degenerate boron doping for TCF reductions. It offers a fast doping based compensation for a $20 \mu\text{m}$ thick SiBAR with TCF reduction to as low as $-2.7 \text{ ppm}/^{\circ}\text{C}$, as shown in Fig. 20(b). Similar to degenerate boron doping, degenerate phosphorous doping has also been demonstrated to reduce TCF in thermally actuated silicon resonators [82].

3) *Geometry Engineering*: Beside composition design and material engineering using doping, geometry engineering is another method in passive temperature compensation for MEMS resonators.

One of earliest geometric stress-compensation designs was demonstrated on low-frequency folded-beam MEMS

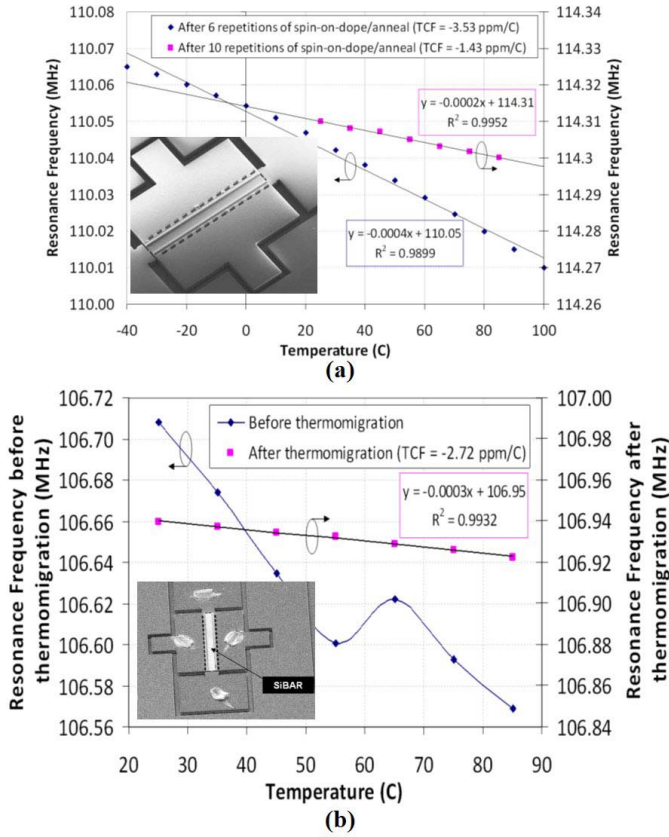


Fig. 20. Degenerate doping for temperature compensation of MEMS resonators. (a) TCF measurements of a $5\ \mu\text{m}$ thick SiBAR using degenerate boron doping with boron density of $\sim 2 \times 10^{20}\ \text{atoms}/\text{cm}^3$, (b) TCF measurements of a $20\ \mu\text{m}$ thick SiBAR with boron-assisted aluminum doping. Reprinted with permission from [195].

resonators in 1998 [202]. Temperature-dependent stresses are introduced through resonator geometry design to compensate the elasticity modulus temperature dependence, achieving TCF reduction for given MEMS resonators. A similar geometric stress-compensation approach was implemented to mechanically compensate the temperature-induced frequency shifts for a high frequency lateral flexural-mode resonator [203]. With this design, a 10 MHz MEMS resonator achieves a large TCF reduction, from $-35.9\ \text{ppm}/^\circ\text{C}$ to $-2.5\ \text{ppm}/^\circ\text{C}$ over a temperature range of $-27\ ^\circ\text{C}$ to $107\ ^\circ\text{C}$, as shown in Fig. 21.

Geometry modifications to the resonator body and its actuation electrodes were also adopted to minimize the TCF of a MEMS resonator. A concave-shaped silicon bulk acoustic resonator (CBAR) was proposed for TCF reduction [204], as illustrated in Fig. 22. Compared to a common silicon bulk acoustic resonator (SiBAR) with a long rectangular geometry, the excitation of a resonance mode with large energy concentration in shear acoustic fields is considered to result in the TCF reduction in CBAR geometry [204]. The temperature compensated CBAR shows a measured TCF of $-6.31\ \text{ppm}/^\circ\text{C}$, compared to $-21.46\ \text{ppm}/^\circ\text{C}$ for a SiBAR fabricated on the same wafer. On the other hand, CBARs compromise the insertion loss and introduce additional spurious modes.

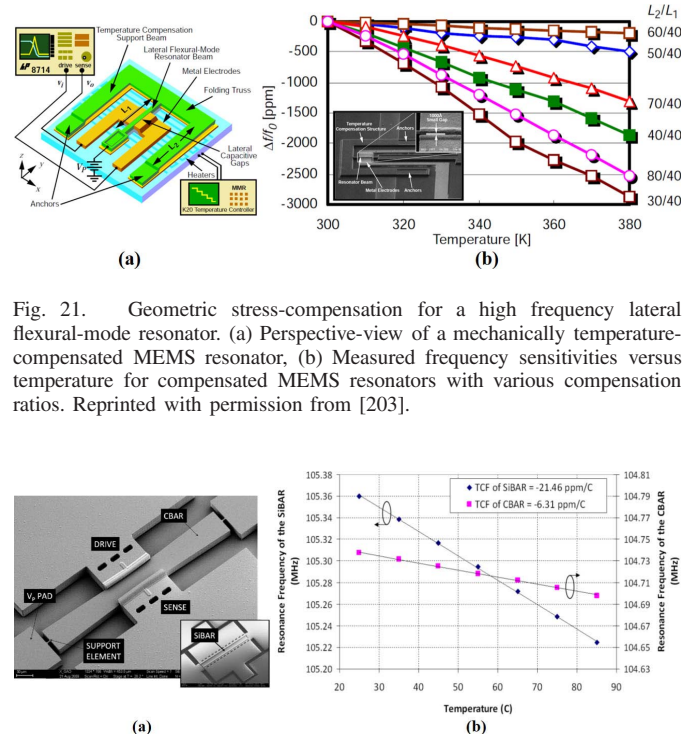


Fig. 21. Geometric stress-compensation for a high frequency lateral flexural-mode resonator. (a) Perspective-view of a mechanically temperature-compensated MEMS resonator, (b) Measured frequency sensitivities versus temperature for compensated MEMS resonators with various temperature compensation ratios. Reprinted with permission from [203].

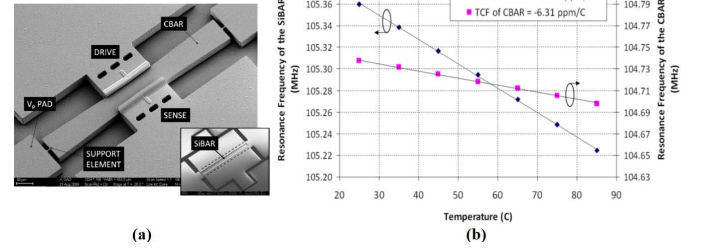


Fig. 22. Geometry modifications of MEMS resonator for TCF reduction. (a) SEM images of a TCF compensated CBAR and a common SiBAR, (b) Comparison of the measured TCF curves of a TCF compensated CBAR and a common SiBAR fabricated on the same wafer. Reprinted with permission from [204].

Resonator arraying is another approach for passive temperature compensation of MEMS resonators [209]. An array of similar resonators are linked to a unique oscillator. The frequency shifts can be compensated by switching from one resonator to the other. This method can also be used for frequency trimming of MEMS resonators.

The active compensation approaches are based on electrically tuning or ovenization of the MEMS resonators. Electrostatic tuning in a feedback loop [182], [210]–[212], electronic compensation via phase-locked loop (PLL) combined with a temperature sensor [11], [16], [89], [102], [213]–[216], thermally isolated micro oven control [217]–[227], are usually used as the active temperature compensation methods for MEMS resonators.

4) *Electrostatic Tuning*: Electrostatic tuning is based on the mechanism of the electrostatic stiffness tuning via a bias voltage for capacitive MEMS devices. A feasibility study of electrostatic tuning for temperature compensation of capacitive MEMS resonators was demonstrated in 2005 [182], [210]. An automatic temperature dependent bias voltage was applied to the resonator body, which compensated the frequency shifts caused by temperature variations [182], [210]–[212], as shown in Fig. 23. The compensated oscillator exhibited a frequency sensitivity of $39\ \text{ppm}$ over the temperature range of $25\ ^\circ\text{C}$ to $125\ ^\circ\text{C}$, which represented a reduction of frequency sensitivity over $70\times$ over the uncompensated oscillator. It is worth to noting that large bias voltages are required in the electrostatic tuning approach, which is conventionally not available in standard CMOS.

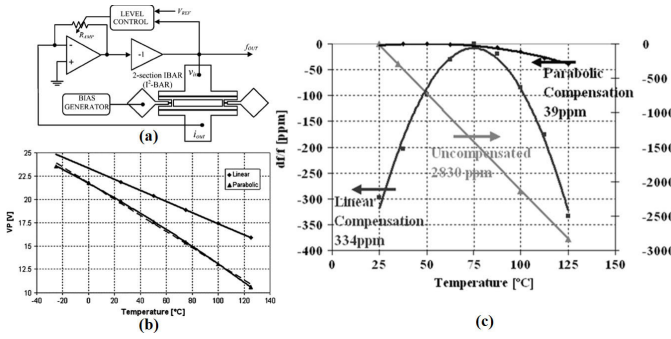


Fig. 23. Temperature compensation for MEMS resonator-based oscillator using electrostatic tuning. (a) Block diagram of a reference oscillator, in which a temperature-dependent bias voltage is applied to resonator body, (b) Generated bias voltage versus temperature for linear and parabolic compensation circuits, and (c) Comparison of measured temperature sensitivities of the oscillator with linear, parabolic compensation and without compensation. Reprinted with permission from [211].

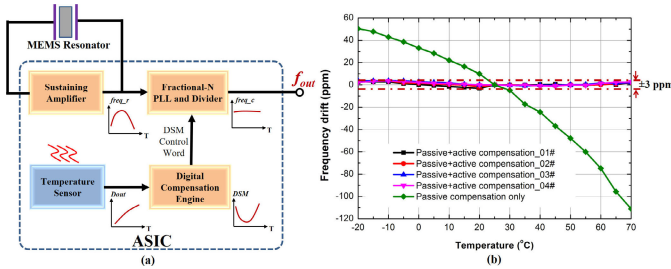


Fig. 24. Active temperature compensation for a piezoelectric MEMS oscillator using electronic compensation. (a) Block diagram of the piezoelectric MEMS oscillator. A temperature sensor and a digital compensation engine are employed to implement the active temperature compensation using a third-order polynomial correction. (b) Comparison of the measured output frequency sensitivities of the MEMS oscillator before and after the active temperature compensation in temperature range between -20°C and $+70^{\circ}\text{C}$. Reprinted with permission from [11].

5) *Electronic Compensation*: Electronic temperature compensation for MEMS resonators is usually implemented via a PLL combined with a temperature sensor [213]–[216]. The compensated frequency stability is mainly determined by the temperature resolution that the temperature sensor can achieve. A major drawback of this approach is the increased phase noise and jitter of the output clock due to the adoption of the PLL circuit.

A PLL-based temperature compensation for a flexural mode MEMS resonator was reported in 2006 [213]. The compensated MEMS resonator shows a total frequency change between $+5$ ppm and -3 ppm over -40°C to $+85^{\circ}\text{C}$.

An active temperature compensation for a piezoelectric MEMS oscillator using electronic compensation has been reported in [11]. Figure 24(a) shows the block diagram of the piezoelectric MEMS oscillator. The ambient temperature is measured with an on-chip resistive temperature sensor embedded in the ASIC chip. The digital compensation engine provides a proper control word to the fractional-N synthesizer, based on a digital third order polynomial correction. After the active compensation, the frequency sensitivities of four MEMS oscillators are reduced to less than ± 3 ppm over the temperature range of -20°C to $+70^{\circ}\text{C}$, as shown in Fig. 24(b).

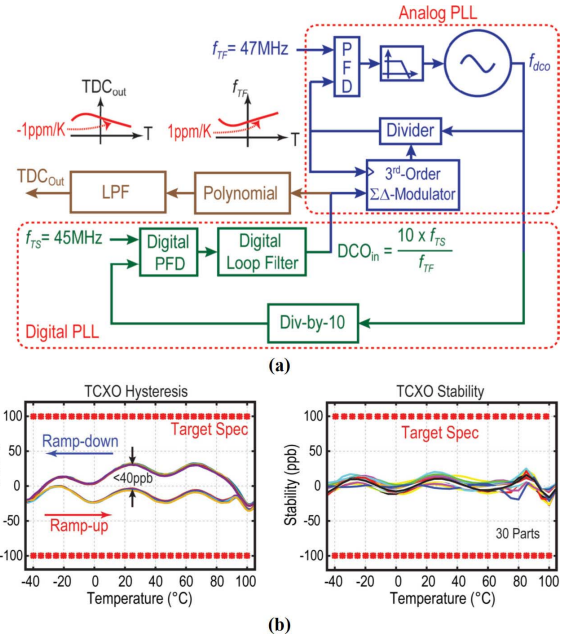


Fig. 25. Dual-MEMS-resonator based temperature sensor with high resolution for high grade electronic temperature compensation. (a) Block diagram of the dual-MEMS-resonator based temperature sensor. The temperature is obtained by measuring the ratio of the frequencies of two oscillators with different $TCFs$. (b) Measured output clock frequency stability, as well as the hysteresis test result for the output clock frequency of a temperature compensated device based on the high resolution temperature sensor. Reprinted with permission from [228].

In order to further improve the temperature resolution, and thus the frequency stability, for electronic temperature compensations of MEMS resonators, a dual-MEMS resonator-based temperature sensor with $20\text{-}\mu\text{K}$ resolution was developed [228]. As shown in Fig. 25(a), the temperature is sensed by measuring the frequency ratio of two MEMS oscillators with different $TCFs$. The high resolution temperature sensor enables to achieve a MEMS-based programmable oscillator with frequency stability of less than ± 0.1 ppm over the temperature range from -45°C to $+105^{\circ}\text{C}$, as shown in Fig. 25(b).

6) *Micro-Oven Control*: Similar to the technique used in oven-controlled quartz oscillators (OCXO) [3], micro-oven control is based on a heater for Joule heating a thermal-isolated micro-oven, where MEMS resonators are placed inside. The oven temperature is precisely controlled at a temperature that matches the turning point of the frequency versus temperature curve of the resonator. Usually, the temperature of the resonator inside the micro-oven is sensed with a resistive feedback circuit [222], [223], [225], [226]. More accurate temperature sensing could be implemented either by correlating the resonator's Q to temperature [229], or by tracking the frequency difference between two resonators with different $TCFs$ or dual-modes in a resonator with different $TCFs$ [24], [25], [217].

An early demonstration of temperature compensation using micro-oven control was by suspending a MEMS resonator above an integrated micro-oven in early 1990s [219]. A micro-oven consisting of a temperature-regulated encapsulation structure was proposed to directly heat

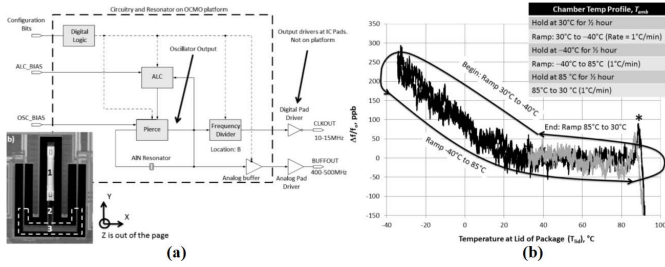


Fig. 26. A fully integrated oven controlled MEMS oscillator. (a) Functional schematic of the OCMO and the picture of released OCMO device. Reprinted with permission from [222]. (b) Measured frequency variations of the OCMO over temperature. Reprinted with permission from [223].

silicon MEMS resonators within a wafer-scale encapsulation, in order to achieve active temperature compensation over the temperature range interested [220].

A fully integrated oven-controlled MEMS oscillator (OCMO) with high thermal isolation and monolithic integration of both MEMS resonators and integrated circuits was reported [222], [223]. The OCMO was achieved based on a AlN on CMOS circuit platform with post-CMOS process. A constant resistance feedback circuit, incorporating with a resistive heater and temperature sensors, was used to sense and stabilize the temperature of the OCMO platform. The OCMO demonstrated measured temperature variations of ~ 300 ppb over the industrial temperature range of -40 °C to 85 °C and ~ 100 ppb over the temperature range of 0 °C to 85 °C with a power consumption of 31 mW, as shown in Fig. 26.

Recently, a dual-mode ovenized MEMS resonator with high temperature stability was reported, in which in-chip integrated micro-ovens are utilized to heat the resonators remaining a stabilized temperature [227]. The resonator operates with two resonance modes simultaneously in one resonator body and is anchored through an in-plane micro-oven suspension frame. Dual-mode operation allows to accurately monitor the resonator temperature and generate a high precise control signal for the temperature control loop. Figure 27(a) shows the mode shapes of a dual-mode ovenized Lamé resonator and Fig. 27(b) illustrates the schematic of experimental set up for the dual Lamé mode resonator. The ovenized Lamé mode resonator achieves a 1-week frequency stability near 1.5 ppb over temperature range from -40 °C to $+60$ °C, as shown in Fig. 27(c).

A combination of passive and active temperature compensation approaches can greatly improve the frequency-temperature stability of MEMS resonators. Passive approaches, such as composite structure and heavy doping, can realize largely reduction of the initial $TCFs$ of the MEMS resonators. Then, active approaches, such as electronic compensation and micro-ovenization, can be used to further tune the frequency stability, in order to achieve better frequency-temperature stability.

D. Initial Frequency Uniformity

In frequency reference and time applications, as the frequency-selective element, the frequency of MEMS

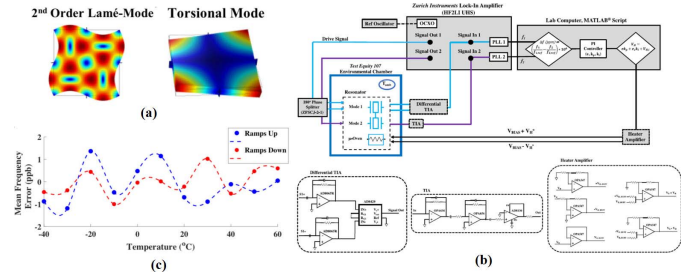


Fig. 27. Low power dual-mode ovenized MEMS resonators based on Epi-Seal platform. (a) Mode shapes of a dual-mode ovenized Lamé resonator, (b) Schematic of experimental set up for the dual Lamé mode resonator, and (c) Measured averaged frequency errors for the 2nd order Lamé mode resonator at steady state with changes in ambient temperature, respectively. Reprinted with permission from [227].

resonator should not only be accurate, but also be precise or uniform. In other words, the resonators should have a uniform pre-defined or targeted resonant frequency. The uniformity of the targeted frequency is limited by manufacturing tolerances in the fabrication processes.

The variations in the dimensions introduced by the tolerances inherent to the fabrication processes like photolithography and etching cause the frequencies of the MEMS resonators to deviate from the designed target values. The natural imperfection in the fabrication processes like film deposition causes the non-uniformities of film thickness, material property, and residual stresses of the deposited film, which results in the frequency non-uniformity of the MEMS resonators across die to die, wafer to wafer, lot to lot, and batch to batch.

Careful design of MEMS resonator makes it possible to reduce the frequency variations due to process tolerances [230]–[232]. In this approach, the effective mass and the effective stiffness have the same proportional variations, resulting in less change of the resonant frequency, as expressed in Eq. (2). However, this approach has some design limitations for some resonators, in which resonator structures cannot have release holes or long tether beams. Besides, the packaging stress-induced frequency shift is also not accounted for in this approach.

Frequency precision of below 100 ppm is required in most applications [1]. In order to improve the frequency uniformity, post-fabrication trimming is usually used to tune the initial frequency of the MEMS resonators to the target values with high accuracy. Commonly, the initial frequency uniformity of MEMS resonators can be improved either by mechanical trimming or electrical trimming.

In mechanical trimming, the mass or stiffness of the MEMS resonators are changed via mass loading or laser ablation to change the resonant frequency [233]–[236]. The frequency variations are in-line monitored during the trimming process and tuned to the targeted frequency with high accuracy. Selective addition of material such as using metal deposition, partial removal of material such as using laser ablation, and material property management such as stress tuning are the most commonly used methods for mechanical trimming.

1) *Mass and Stiffness Tuning*: The resonant frequency of a MEMS resonator is dependent on its effective mass and

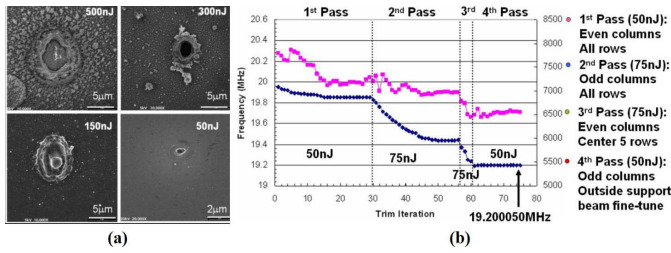


Fig. 28. Frequency trimming for MEMS resonators using laser ablation. (a) Laser spot characterization on polysilicon with the energy ranging from 500 nJ down to 50 nJ, and (b) Measured frequencies of the laser trimmed MEMS resonator down to from 4% down to 2.6 ppm. Reprinted with permission from [234].

stiffness. Trimming with laser or ion beam, is a cost-effective method to perform post-fabrication frequency trimming of MEMS resonators [237], [238]. Laser or ion beam trimming can not only tune the mass, but also modify the stiffness of the MEMS resonators, achieving a bidirectional frequency trimming.

One of earliest works for frequency tuning of MEMS resonators using laser trimming was reported in 2004 [233]. The trimmed resonators demonstrate an impressive initial frequency error reduction of from 2% to 21 ppm. Packaged MEMS resonators can also be trimmed with laser, as long as the cap of resonator package is transparent to laser beam. The trimming accuracy can be controlled by tuning the laser power and optimizing the trimming locations [233], [234]. As shown in Fig. 28(a), laser energy has large effect on the trimming accuracy. Using a high laser power will have large trimming capability, and is suitable for coarse tuning at the beginning. In contrast, low laser power will have small frequency trimming capability but higher resolution, and is suitable for fine tuning. A frequency accuracy of 2.6 ppm has been achieved with laser trimming [234], as shown in Fig. 28(b).

Pulsed laser deposition (PLD) is also adopted as a post-packaging frequency tuning method for MEMS resonators [235]. Donor material, such as gold, indium or aluminum is added on the structure surface of a MEMS resonator using PLD, resulting in frequency change and achieving the frequency tuning for packaged MEMS resonators. However, the frequency accuracy is not high, due to the poor control of mass variation.

2) *Stress Tuning*: Frequency trimming can also be realized by changing the residual stress of the deposited film on the resonator using rapid thermal annealing (RTA) approach [239]. The stress change due to RTA will result in the variation of the effective modulus, and thus tune the resonant frequency of the MEMS resonator. The frequency variations are highly linear with the annealing temperature using this trimming approach. The measured results of a wafer level packaged AlN MEMS resonator show that the frequency variation across-wafer is reduced from 22000 ppm to 4000 ppm.

In electrical trimming, the effective stiffness, effective mass or material properties are trimmed electrically, such as bias tuning [212], [234], eutectic alloying via electrically heating [240]–[242], micro-evaporator [243], in order to tune

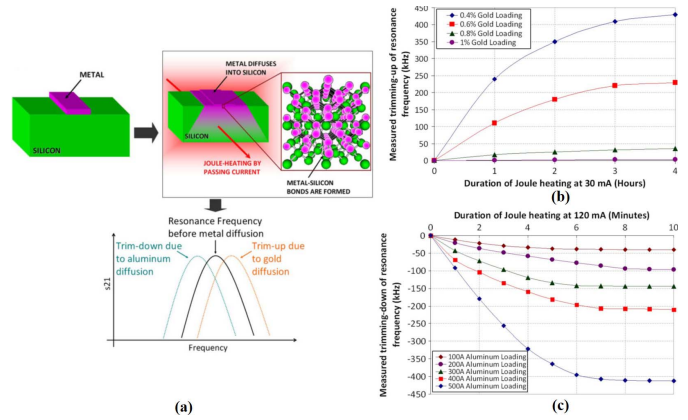


Fig. 29. Post-fabrication electrical trimming of silicon MEMS resonators using metal thermal diffusion via Joule heating. (a) Schematic of the metal thermal diffusion into silicon via Joule heating and its application to frequency trimming of MEMS resonators. (b) Measured trimming-up of gold-loaded SiBARs and (c) trimming-down of aluminum-loaded SiBARs. Reprinted with permission from [241].

the resonator's frequency. Electronic frequency trimming via PLL is another effective frequency trimming method with high accuracy [213], [214], [234]. Electronic trimming is also widely used in the temperature compensation for MEMS resonators, as described in Section IV-C.

3) *Bias Tuning*: Similar to the approach used for the frequency temperature compensation of MEMS resonators described in Section IV-C, the resonator's frequency can be tuned by its bias voltage using the electrostatic spring softening phenomenon [212], [234]. With the aid of the PLL-based frequency synthesizer, the frequency trimming can achieve high accuracy below ppm level. However, dedicated circuits are required and the tuning range with this approach is dependent on the bias voltage range provided by the circuit. PLL also degrades the phase noise performance of the MEMS devices.

4) *Material Properties Controlling*: Frequency trimming can be also achieved by controlling the material properties of the resonator structure, either using electrically, optically or thermally approaches.

A post-fabrication electrical trimming of silicon MEMS resonators using metal thermal diffusion into silicon via Joule heating has been reported in [240], [241]. Figure 29(a) illustrates the schematic of the metal thermal diffusion into silicon via Joule heating and its application to frequency trimming of MEMS resonators. Gold or aluminum thermal diffusion into bulk silicon of the resonant body is used to trim up or trim down the resonant frequency via Joule heating, respectively. Bidirectional trimming by ~ 4000 ppm has been demonstrated using gold and aluminum diffusion in 100 MHz SiBARs via Joule heating, as shown in Fig. 29(b) and Fig. 29(c), respectively. However, the residual metal film on the device surface may deteriorate device's Q and performance.

Localized thermal oxidation via electronically heating is an alternative way for frequency trimming of MEMS resonators [244], [245]. As shown in Fig. 30(a) and (b), an I-shaped SCS resonator body is oxidized by extreme

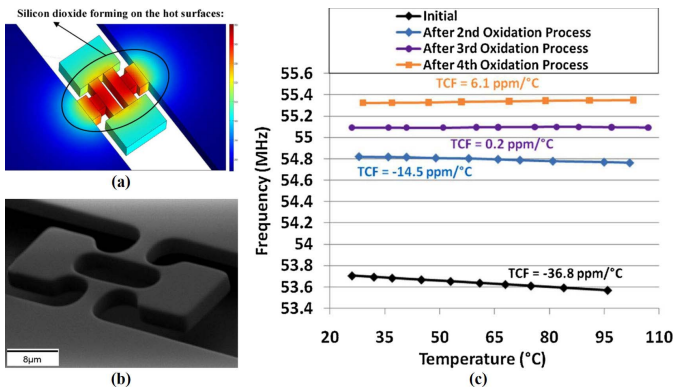


Fig. 30. Electronically controllable frequency trimming technique based on localized thermal oxidation of SCS MEMS resonators. (a) Schematic view of MEMS resonator heating up with a large biased dc current and hot surface is oxidized to form SiO_2 . (b) SEM image of the reported I-shaped MEMS resonator, and (c) Measured frequency versus temperature curves for a trimmed MEMS resonator with SiO_2 on the surface of the resonator. Reprinted with permission from [245].

Joule heating using large bias currents, and thus the effective stiffness of the resonator is changed, resulting in shifts of the resonant frequency. Both initial frequency trimming and frequency temperature compensation can be realized using this technology. Figure 30(c) shows the measured frequency versus temperature curves for a 54 MHz I-shaped MEMS resonator with SiO_2 on the surface of the resonator. The trimmed resonator demonstrates a frequency trimming range of $\sim 3.7\%$ and a TCF of $0.2 \text{ ppm}/^\circ\text{C}$, respectively.

Recently, another frequency trimming method for MEMS resonators by excimer laser annealing of Ge thin film on silicon has been reported in [242]. Focused laser beam is used to locally heat up Ge thin film to high temperatures, enabling selective micro-crystallization of SiGe, and thus change the structure stiffness of MEMS resonators. A large trimming range over 5000 ppm with this technology has been demonstrated for a low-frequency out-of-plane mode square resonator. The trimmed resonator remains stable after high temperature annealing and no obvious damping is introduced.

5) *Integrated Micro-Evaporator*: An alternative method for post-packaging frequency trimming of the MEMS resonators has been reported, which is based on an integrated micro-evaporator to deposit the donor material, such as aluminum, on the resonator surface [243]. The micro-evaporator is heated up using integrated heaters to the saturation vapor pressure of the donor material. The donor material is evaporated and deposited on the resonator surface, thus shifting the resonator's frequency. The average trimming rate is largely dependent on the electric power applied to the heater. Both coarse and the fine trimming can be realized using this approach.

V. BUSINESS ACTIVITIES

MEMS resonators demonstrate several benefits over quartz crystal resonators, such as semiconductor processes and packaging, less sensitivity to shock and vibration, compact size and low power. The long-term stability of MEMS oscillators has shown competitive performance with the quartz crystal

oscillators, in the sub-ppm range. Numerous MEMS resonators have been commercialized and are in mass production since the early 2000s, by a dozen of startups, including SiTime, Discera, Silicon Clocks, Sand9, Tikitin, etc.

SiTime and Discera are the top two players in MEMS timing devices market. Both of them are developing capacitive MEMS timing devices. Silicon Clocks was founded in 2004 and acquired by Silicon Labs in 2010. Silicon Clocks had an innovative CMEMS technology, which built poly-SiGe based MEMS devices on top of CMOS wafers, achieving MEMS and CMOS monolithic integrations [136], [246]. Sand9 was founded in 2007 and acquired by Analog Devices in 2010. Sand9's piezoelectric MEMS resonators can be integrated into other semiconductor component devices for mobile and consumer electronics. Tikitin, founded in 2016, is a latest spin-off from Finnish research institute VTT, which offers both capacitive and piezoelectric MEMS devices. However some of the startups have already quietly removed the MEMS timing devices from their product catalogs, for various development, marketing and capitalization problems.

Epson is one of the major players in quartz crystal devices, but also in quartz MEMS (QMEMS) devices. QMEMS is a combination of the high stability and precision of quartz and MEMS microfabrication technology [247]. It brings a big evolution for the fabrication process of conventional quartz timing devices and makes the quartz business more promising again. Murata is a new player in MEMS resonator market. Murata has developed and released a small size, low power consumption, 32.768 kHz piezoelectric MEMS resonator in 2018, in which the load capacitors typically used in quartz crystal devices are also embedded in the package. Recently, Texas Instruments also started to develop MEMS timing devices based on a bulk acoustic wave (BAW) resonator technology in 2019 [42], [164], which when integrated into the MCU package eliminate the need for external quartz crystals without compromising power, latency or frequency stability.

A. SiTime

SiTime was founded in 2004, acquired by MegaChips in 2014, and became a free-standing company starting in November 2019. SiTime had over 90 percent of the market in 2016 and is still the leader in the MEMS timing devices.

There are two basic MEMS resonator designs in SiTime [248], [249]: 524 kHz H-style tuning-fork, and 48 MHz four-ring bulk mode resonators, as shown in Fig. 31. Both of the two resonators are fixed at the resonator center with single anchors. The single-anchored design can largely decouple the packaging induced stresses and has less sensitive to external stresses, which benefit a lot to the long term stability of the MEMS resonators.

The MEMS resonators are fabricated and packaged using SiTime's MEMS-First and Epi-Seal process [120], [249]. The high temperature vacuum sealing technology provides a clean and stable environment, and thus ensure high Q and good long term stability for MEMS resonators.

DualMEMS high resolution temperature sensing with TurboCompensation enables SiTime's MEMS oscillators

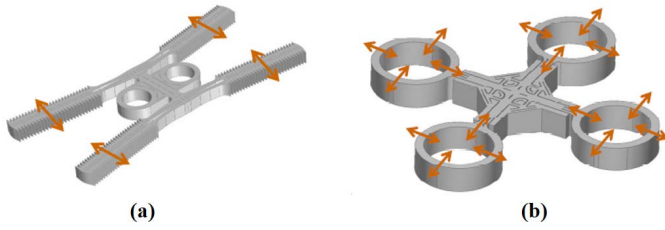


Fig. 31. Two basic types of MEMS resonators in SiTime. (a) 525 kHz resonator and (b) 48 MHz resonator. Reprinted with permission from [249].

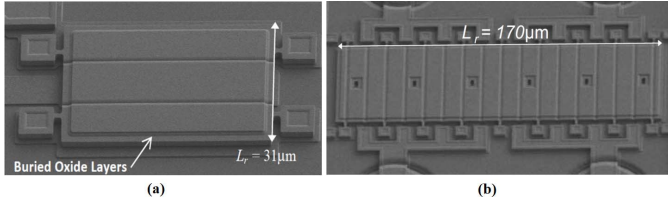


Fig. 32. SEM images of two basic types of MEMS resonators in Microchip. (a) Free-free beam resonator and (b) 15th overtone free-free beam resonator. Reprinted with permission from [90].

demonstrating high frequency stability of less than ± 0.1 ppm over the temperature range from -45 °C to $+105$ °C [228], [248]. SiTime has also developed OCMOs based on their Emerald platform. Emerald OCXOs provide the best dynamic performance, with frequency stability of less than ± 5 ppb over temperature range from -40 °C to $+85$ °C [228], [248].

B. Microchip

Microchip acquired MEMS timing technology through the purchases of Discera and Micrel in 2015. Discera was the first MEMS timing devices startup company, which spun off from University of Michigan, and was founded in 2001. Discera was bought by Micrel in 2013, which was then acquired by Microchip in 2015.

Different from SiTime's, the resonators developed by Microchip operate on out-of-plane flexural mode, which was fabricated by poly-silicon based surface micromachining process [90], [125], [135]. Figure 32 shows the SEM images of a free-free beam resonator operating in the fundamental flexural mode, and a 15th overtone flexural mode resonator, which are two typical MEMS resonator designs in Microchip [90], [135]. After fabrication, the MEMS resonators are hermetically sealed by a wafer level vacuum bonding approach. Glass frit and fusion bonding were adopted in the first generation and later generation products, respectively. The fusion bonding makes the resonator more robust, improves the yield and largely reduces the size. Microchip's oscillators show average drift of around -1.5 ppm under constant operation at $+85$ °C for 1,000 hours.

C. Tikitin

The mission of Tikitin is to replace quartz crystal in time and frequency control applications with MEMS based devices. Tikitin uses proprietary doping technology to implement the

temperature compensation for MEMS resonators. A narrow ± 25 ppm frequency variation over a wide temperature range from -40 °C to $+150$ °C has been achieved. HighQ and highT are two basic MEMS resonator technologies at Tikitin. HighQ uses capacitive MEMS resonators, usually operating in with bulk modes, which show high Q values of around 1 million. HighQ resonators are ideal for the high-precision frequency control applications. HighT uses thin-film AlN on silicon piezoelectric MEMS resonators, which has low R_m of around 100 Ohm. HighT resonators are expected to be the best candidate for pin-to-pin replacement with quartz devices. Tikitin was purchased by Kyocera in 2019, and are preparing for mass producing silicon MEMS resonators.

VI. TRENDS AND OPPORTUNITIES

MEMS timing devices have the potential to replace quartz crystal based timing solutions. The future development trends of MEMS resonators are still high performance, low cost, and high integration level, either driven by technologies, market, or applications.

A. Technology-Driven Trends

Technology is one of the most important driving forces for MEMS resonators. MEMS resonators have experienced a long development period and they still need much effort to make them better than quartz. With the technological advancements in the MEMS, MEMS resonators have been rapidly growing from being inferior to rivaling quartz resonators. Epi-Seal and fusion bonding processes enable a high stability, compact size and mass producible packaging for MEMS resonators. The innovative on-chip temperature sensing technology achieves compensated MEMS resonators with high frequency stability over temperature.

In the future, advancements in materials may greatly drive the developments of MEMS resonators. Degenerated homogeneous doping could improve the $TCFs$ uniformity of the MEMS devices, resulting in better performance and higher yield. For piezoelectric MEMS resonators, doped AlN or single crystal AlN could be adopted in order to enhance the electromechanical coupling factor. Moreover, single crystal AlN has better crystal orientation and lower acoustic damping, compared to the commonly used orientation-preferred poly-crystal AlN, resulting in a higher Q and better stability.

CMOS-MEMS technologies could also offer much space for the development of MEMS resonators. Several CMOS circuit technologies include materials such as GaN for RF applications [250], or PZT for ferroelectric memory [251]. These materials are piezoelectric as well and such CMOS technologies may be leveraged directly or with minimal modifications to enable high quality MEMS resonators to be built within the process. Advanced-node CMOS may also be leveraged for electrostatic MEMS resonators with unreleased acoustic cavities to be developed within the process, adjacent to CMOS [252]–[254]. The tight integration of CMOS and MEMS will minimize parasitics and power consumption, greatly reduce the system size and cost, and achieve integration of multi-functional devices on a single chip.

B. Market-Driven Trends

Quartz crystal resonators and timing devices are still dominating the overall huge timing market. The main objective for developing MEMS resonators and its timing devices, of course, is to replace the quartz crystal counterparts. As a candidate, MEMS resonators and timing devices are required to have rival or better performance and lower cost, compared to these existing quartz crystal counterparts. There are two possible business models for MEMS timing devices from the market aspect views.

One way is to replace the quartz crystal resonators directly using MEMS resonators, referring to “element” or drop-in replacement for quartz, while remaining the same driven circuit as quartz. The MEMS resonators are required to have much comparable or even a bit better performance, while a lower price than quartz crystal resonators. Piezoelectric MEMS resonators make this conception to be possible. Tikitin has already claimed that their piezoelectric MEMS resonators could directly replace quartz crystal resonators, allowing a pin-compatible replacement.

The other way is providing a MEMS based timing solution based on the MEMS resonators and its dedicated circuits, as a “module-level” replacement. The performance of the MEMS resonators could be different with the quartz crystal resonators. However, with the aid of dedicated circuits, such as PLL combined with on-chip temperature sensors, the overall performance of the MEMS oscillator or MEMS based timing devices could be better than quartz crystal counterparts. Some companies like SiTime offer MEMS oscillator and timing products with industry-standard package to replace the quartz timing devices. This kind of “module-level” replacement is also pin-compatible and no board design changes are necessary in a higher circuit level.

C. Application-Driven Trends

Compact size, low cost and multi-functionalities are driving new frontiers in the electronic circuits of modern mobile devices, especially in the transceiver circuits for cell phones. With the technology trends of rapidly increasing of the integration of electronic systems, more and more timing sources are needed to provide the frequency or time reference for the corresponding function blocks, such as application processor, global navigation satellite system (GNSS) receiver, Wifi and bluetooth IC, camera modules. Advanced packaging technologies, co-packaging or system-in-packaging (SiP) provides an effective way to save the space and lower down the cost. However, currently the typically used timing sources are quartz crystal frequency references. Quartz crystal oscillators are packaged either in metal or ceramic carriers as discrete components, which does not support their integration or co-packaging with other devices or modules.

MEMS resonators and oscillators feature small sizes, and are fabricated and packaged using semiconductor processes, which are a promising alternative to be integrated or co-packaged with other circuits or modules in one packaging. Therefore, MEMS resonators and oscillators have unique advantages over the quartz crystal oscillators in above

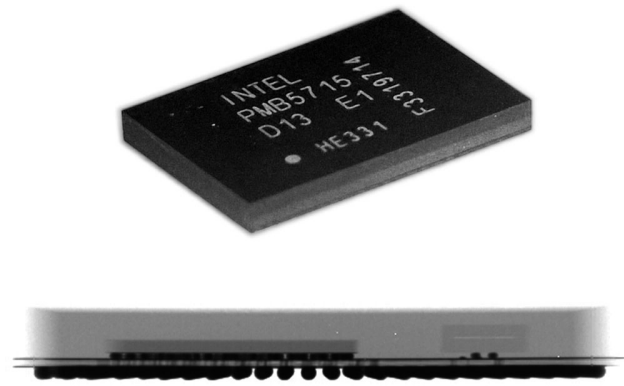


Fig. 33. Transceiver with integrated MEMS resonator. The top is the image of the transceiver, and the bottom shows the X-ray cross-sectional view of the packaging. The transceiver IC and MEMS resonator chip are co-packaged side by side using SiP technology. Reprinted with permission from [106].

mentioned applications, where higher level integration and space saving are becoming important. A first implementation of a SiP cellular transceiver with integrated frequency reference was reported in [106]. A MEMS resonator was integrated into the same package of a processor and provided timing signals to the processor, as shown in Fig. 33. This demonstration shows a typical example of new development trends for MEMS resonators, which is mainly driven by the requirements of the applications.

Low jitter clock is another application area of MEMS timing solutions. MEMS timing devices that do not require a PLL have a distinct advantage over lower frequency quartz crystal timing products, which must be frequency up-converted using a PLL circuit, causing the PLL circuit with its low Q tank to dominate the jitter. Texas Instruments’s BAW technology uses a high frequency, high Q BAW MEMS resonator co-packaged with silicon IC to implement an ultra-low noise integrated voltage-controlled BAW oscillator (VCBO), which greatly reduces the synchronized output clock jitter [42].

VII. OUTLOOK

With the rapid development of MEMS technologies, MEMS resonators are becoming more and more promising to replace the quartz crystal counterparts in frequency reference and timing applications. In the future, SiP with other modules, multi-frequency functionality in a single chip, and high stability are three major technology driving forces for MEMS resonators.

SiP allows heterogenous integration and offers smaller package size and lower I/O count, which is one of most important technologies demanded across various applications, as described in Section VI. However, quartz crystal resonators cannot be SiP. One of the factors is attributed to the packaging form using ceramic or metal carrier of quartz crystal resonators, resulting in large packaging size. The most important reason is the high failure rate of quartz crystal resonators [17], [18], which hinders their integration with other modules. MEMS resonators are made using semiconductor process with much better process control, hence eliminating the issues on

the quartz resonators. Compared to quartz crystal resonators, MEMS resonators demonstrate significantly (up to 80×) better reliability [18]. Therefore, the high reliability and compact size allow MEMS resonators to co-packaging or SiP with other modules. The SiP technologies will greatly push forward the developments of MEMS resonators in the next following years.

With rapid growth of the demands on wireless communications, multi-band and multi-frequency are required in the system. However, quartz crystal resonators can not achieve multi-frequencies in a single chip, since the fabrication line of quartz crystal resonators is dedicated to a product with fixed design and hence fixed resonant frequency. MEMS resonators are fabricated on wafer-level using semiconductor process, which allows to integrate multi-frequency resonators in a single chip with the same fabrication process. Thus, multi-frequency functionality in a single chip grants MEMS resonators more space in the timing markets in future.

High stability, of course, is most important factor for MEMS resonators to gain a competitive edge over quartz crystal resonators. Currently, MEMS resonators have demonstrated comparable performance with the typical quartz counterparts. Especially, MEMS resonators have shown excellent performance and applied in high-end applications, such as automotive, industrial GNSS, and satellite [248]. However, the initial frequency uniformity and *TCF* uniformity of MEMS resonators are still inferior to the quartz crystal resonators. Much effort is needed, in terms of careful design, material quality control or new material development, precise process control, and packaging, in order to improve MEMS resonator performance. While circuit-level approaches like ASIC and compensation algorithms will complement the device-level efforts. MEMS resonators will have much room to further improve their performance for high-end applications.

VIII. CONCLUSION

This paper provides a comprehensive overview on the resonator design, fabrication process, existing key technologies for improving performance, business activities, future trends and the opportunities of MEMS resonators.

The mechanical and electrical models of MEMS resonators are discussed. Various vibration modes are illustrated and compared. Two main transduction types of MEMS resonators, capacitive and piezoelectric, are addressed in detail. Commonly used fabrication technologies, including surface micromachining, SOI/CSOI based processes, HARPSS technologies, and TPoS platform, are reviewed. Fusion and Al-Ge eutectic bonding, as well as the Epi-Seal process are the widely used wafer-level packaging approaches for MEMS resonators.

Key technologies for improving the major performance of MEMS resonators, such as Q , motional impedance, temperature stability, and frequency uniformity are reviewed in detail. To reduce anchor losses, anchoring at nodal points, quarter wavelength tethers, material-mismatched supports, acoustic reflection cavities, and phonon crystal tethers are commonly used approaches. Thermoelastic damping can be reduced using geometry design and device structure optimization. Motional impedance could be reduced by manipulation of the electro-mechanical coupling factor or using array designs. Passive and

active temperature compensations are implemented to enhance the temperature stability of MEMS resonators. Initial frequency uniformity could be improved by mass/stiffness tuning, stress tuning, electrical tuning or controlling material properties. The business activities and major players in the MEMS timing market, and future development trends driven either by technology, market, or applications are discussed.

The technologies discussed in this paper can not only be used for achieving high performance MEMS resonators for frequency reference and timing applications, but also be adopted to realize high resolution sensors, as well as high quality RF MEMS filters.

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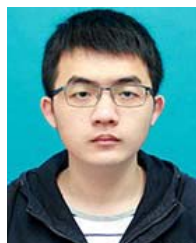
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